blood advances

American Society of Hematology
2021 L Street NW, Suite 900,
Washington, DC 20036
Phone: 202-776-0544 | Fax 202-776-0545
bloodadvances@hematology.org

American Society of Hematology 2026 Guidelines for Immune Thrombocytopenia (ITP): Initial and Second-Line Therapy in Adults with Primary ITP

Tracking no: ADV-2026-021269R1

Adam Cuker (University of Pennsylvania, United States) Deirdra Terrell (University of Oklahoma Health Sciences Center, United States) Honieh Sowdagar (University of Oklahoma Health Sciences Center, United States) Donald Arnold (McMaster University, Canada) Sylvain Audia (Université Bourgogne Europe, CHU Dijon Bourgogne, France) James Bussel (Department of Pediatrics, Division of Hematology/Oncology, Weill Cornell Medicine, United States) Tomas Jose Gonzalez-Lopez (Hematology, Hospital Universitario de Burgos, Burgos, Spain) Rachael Grace (Dana-Farber/Boston Children's Cancer and Blood Disorders Center, United States) Camila Masias (Miami Cancer Institute - Baptist Health South Florida, United States) Keith McCrae (Cleveland Clinic, United States) Mirjana Mitrovic (Univesity Clinical Center of Serbia, Serbia) Cindy Neunert (Columbia University Irving Medical Center, United States) Sandhya Panch (Fred Hutchinson Cancer Center/University of Washington, United States) Surbhi Shah (Mayo clinic Arizona, United States) Brenda Shy (NA, United States) Jessica VandeVelde (NA, United States) Douglas Cines (Perelman School of Medicine, University of Pennsylvania, Philadelphia, Pennsylvania, United States) Sara Vesely (University of Oklahoma Health Sciences Center, United States)

Abstract:
Background: Since publication of the American Society of Hematology 2019 Guidelines for Immune Thrombocytopenia (ITP), new clinical trials have been completed and new treatments have become available. Objective: These evidence-based guidelines from the American Society of Hematology (ASH) are a focused update of the 2019 ASH ITP guidelines, centering on treatment of adults with primary ITP. Methods: ASH formed a multidisciplinary guideline panel including two patient representatives that was balanced to minimize potential bias from conflicts of interest. The University of Oklahoma supported the guideline development process, including updating or performing systematic evidence reviews (up to July 19, 2025). The panel prioritized clinical questions and outcomes according to their importance for clinicians and patients and identification of areas with new data or where application of new methods would be of benefit. The panel used the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach, including GRADE Evidence-to-Decision frameworks and the GRADE framework for multiple comparisons, to assess evidence and make recommendations, which were subject to public comment. Results: The panel agreed on recommendations in adults with primary ITP regarding initial therapy and second-line treatment in patients who failed first-line corticosteroids. The panel also issued good practice statements on thrombopoietic agent switching and splenectomy. Conclusions: Novel recommendations of these guidelines include: (1) initial therapy with a combination of rituximab plus corticosteroids or a thrombopoietic agent plus corticosteroids rather than corticosteroids alone (conditional recommendation) and (2) a thrombopoietic agent (strong recommendation) or rituximab (conditional recommendation) for patients who failed first-line corticosteroids.

Conflict of interest: COI declared - see note

COI notes: COI disclosures are detailed in Supplemental files 2 and 3.

Preprint server: No;

Author contributions and disclosures: AC, DRT, and SKV wrote the first draft of this manuscript and revised the manuscript based on authors' suggestions. Guideline panel members (DMA, SA, JBB, TJG-L, RFG, CM, KRM, MM, CN, SRP, SS, BS, JV) critically reviewed the manuscript and provided suggestions for improvement. Members of the knowledge synthesis team (DRT, HS, SKV) contributed evidence summaries to the guidelines. All authors approved of the content. DBC and SKV were the co-chairs of the panel and led panel meetings.

Non-author contributions and disclosures: No;

Agreement to Share Publication-Related Data and Data Sharing Statement:

Clinical trial registration information (if any):

American Society of Hematology 2026 Guidelines for Immune Thrombocytopenia (ITP): Initial and Second-Line Therapy in Adults with Primary ITP

Adam Cuker[1], Deirdra R. Terrell[2], Honieh Sowdagar[3], Donald M. Arnold[4], Sylvain Audia[5], James B. Bussel[6], Tomás José González-López[7], Rachael F. Grace[8], Camila Masias[9], Keith R. McCrae[10], Mirjana Mitrovic[11], Cindy Neunert[12], Sandhya R. Panch[13], Surbhi Shah[14], Brenda Shy[15], Jessica VandeVelde[16], Douglas B. Cines[17,*], Sara K. Vesely[18]

Corresponding author:

Adam Cuker, MD, MS
Hospital of the University of Pennsylvania
3400 Spruce Street
Philadelphia, PA 19104
Adam.Cuker@pennmedicine.upenn.edu

1. Department of Medicine and Department of Pathology & Laboratory Medicine, Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA, USA; Adam.Cuker@pennmedicine.upenn.edu; ORCID ID 0000-0002-3595-5697
2. Department of Biostatistics and Epidemiology, Hudson College of Public Health, University of Oklahoma Health Campus, Oklahoma City, OK, USA; dee-terrell@ou.edu; ORCID ID 0000-0003-0794-5851
3. Department of Biostatistics and Epidemiology, Hudson College of Public Health, University of Oklahoma Health Campus, Oklahoma City, OK, USA; Honieh-sowdagar@ou.edu
4. Department of Medicine and Michael G. DeGroote Centre for Transfusion Research, McMaster University, Hamilton, ON, Canada; arnold@mcmaster.ca
5. Université Bourgogne Europe, CHU Dijon Bourgogne, Department of Internal Medicine, Dijon, France; sylvain.audia@u-bourgogne.fr; ORCID ID 0000-0003-1772-1392
6. Department of Pediatrics, Weill Cornell Medical College, Cornell University, New York, NY, USA; jbussel@med.cornell.edu
7. Department of Hematology, Hospital Universitario de Burgos, Burgos, Spain; tjgonzalez@saludcastillayleon.es; ORCID ID 0000-0001-9575-1816

8. Boston Children's Cancer and Blood Disorders Center, Department of Pediatrics, Harvard Medical School, Boston, MA, USA; rachael.grace@childrens.harvard.edu; ORCID 0000-0001-7302-0449
9. Miami Cancer Institute – Baptist Health South Florida, Coral Gables, FL, USA; camilamas@baptisthealth.net
10. Department of Hematology and Medical Oncology, Taussig Cancer Institute, Cleveland Clinic, Cleveland, OH, USA; mccraek@ccf.org
11. Clinic of Hematology, University Clinical Center of Serbia and Medical Faculty, University of Belgrade, Belgrade, Serbia; mirjanamitrovic777@gmail.com; ORCID ID 0000-0001-8313-3754
12. Division of Pediatric Hematology, Oncology, and Stem Cell Transplantation, Columbia University Irving Medical Center, New York, NY, USA; cn2401@cumc.columbia.edu
13. Division of Hematology/Oncology, University of Washington/Fred Hutch Cancer Center, Seattle, Seattle, WA, USA; srpanch@uw.edu
14. Division of Hematology and Medical Oncology, Mayo Clinic, Phoenix, AZ, USA; shah.surbhi@mayo.edu
15. Albany, GA, USA; bgshy79@gmail.com
16. Topeka, KS, USA; vandevelde.jess@gmail.com
17. Department or Pathology & Laboratory Medicine, Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA, USA
18. Department of Biostatistics and Epidemiology, Hudson College of Public Health, University of Oklahoma Health Campus, Oklahoma City, OK, USA; sara-vesely@ou.edu; 0000-0003-3448-0156

*Deceased

Abstract

Background: Since publication of the *American Society of Hematology 2019 Guidelines for Immune Thrombocytopenia* (ITP), new clinical trials have been completed and new treatments have become available.

Objective: These evidence-based guidelines from the American Society of Hematology (ASH) are a focused update of the 2019 ASH ITP guidelines, centering on treatment of adults with primary ITP.

Methods: ASH formed a multidisciplinary guideline panel including two patient representatives that was balanced to minimize potential bias from conflicts of interest. The University of Oklahoma supported the guideline development process, including updating or performing systematic evidence reviews (up to July 19, 2025). The panel prioritized clinical questions and outcomes according to their importance for clinicians and patients and identification of areas with new data or where application of new methods would be of benefit. The panel used the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach, including GRADE Evidence-to-Decision frameworks and the GRADE framework for multiple comparisons, to assess evidence and make recommendations, which were subject to public comment.

Results: The panel agreed on recommendations in adults with primary ITP regarding initial therapy and second-line treatment in patients who failed first-line corticosteroids. The panel also issued good practice statements on thrombopoietic agent switching and splenectomy.

Conclusions: Novel recommendations of these guidelines include: (1) initial therapy with a combination of rituximab plus corticosteroids or a thrombopoietic agent plus corticosteroids rather than corticosteroids alone (conditional recommendation) and (2) a thrombopoietic agent (strong recommendation) or rituximab (conditional recommendation) for patients who failed first-line corticosteroids.

Summary of recommendations

These guidelines are based on updated and original systematic reviews of evidence conducted under the direction of the University of Oklahoma. The panel followed best practices for guideline development [1-3]. The panel used the GRADE approach to assess the certainty of the evidence and formulate recommendations[4] .

These guidelines address the management of adults with primary immune thrombocytopenia (ITP). ITP is an acquired disorder characterized by a low platelet count due to autoimmune-mediated platelet destruction and impaired platelet production[5,6]. The estimated incidence of ITP is 1 to 6 per 100,000 adults[7-11]. Common clinical manifestations include bleeding, fatigue, and reduced health-related quality of life[12]. Treatment has evolved in recent years owing to the development of new therapies[13]. In these guidelines, we summarize available evidence and provide recommendations on initial and second-line therapy (needing additional treatment after initial therapy with corticosteroids) in adults with primary ITP. We also provide good practice statements on the role of thrombopoietic agent switching and splenectomy in contemporary practice.

Interpretation of strong and conditional recommendations

The strength of a recommendation is expressed as either strong ("the guideline panel recommends..."), or conditional ("the guideline panel suggests…") and should be interpreted as described in Figure 1.

Interpretation of good practice statements

Good practice statements endorse interventions or practices that the guideline panel agreed have unequivocal net benefit yet may not be widely recognized or used[14]. Good practice statements in these guidelines are not based on a systematic review of available evidence. Nevertheless, they may be interpreted as strong recommendations.

Recommendations

Initial Therapy in Adults with Primary ITP

> **Recommendation 1.** In adults with primary ITP requiring initial therapy, the ASH guideline panel *suggests either* rituximab plus corticosteroids (with or without IVIG) or a thrombopoietic agent plus

corticosteroids (with or without IVIG) rather than mycophenolate mofetil plus corticosteroids (with or without IVIG) or corticosteroids (with or without IVIG) alone [conditional recommendation based on low certainty in the evidence ⊕⊕◯◯ (recommendation limited to rituximab or its biosimilars)].

If rituximab and thrombopoietic agents are not available options, the panel *suggests* corticosteroids (with or without IVIG) alone rather than mycophenolate mofetil (MMF) plus corticosteroids (with or without IVIG) [conditional recommendation based on very low certainty in the evidence ⊕◯◯◯].

Remarks:

- A demonstrable rise in platelet count with corticosteroids and/or IVIG increases confidence in the diagnosis of ITP. Therefore, it is reasonable to treat a patient with a brief course (≤2 weeks) of corticosteroids (with or without IVIG) to increase confidence in the diagnosis of ITP before implementing combination therapy, especially in cases of diagnostic uncertainty.
- Although the most studied corticosteroid component of combination therapy is dexamethasone 40 mg daily × 4 days for 1 to 3 cycles spaced 2 to 4 weeks apart, it is also appropriate to use prednis(ol)one at a starting dose of 0.5-2 mg/kg/day.
- Whether corticosteroids are used as part of combination therapy or alone, it is advisable to limit their administration to 6 weeks or less to minimize toxicity.
- If combination therapy with rituximab is used, the most studied regimen is 375 mg/m2 × 4 weekly doses. However, other doses and schedules of rituximab may be used according to local practice. Biosimilars to rituximab are suitable substitutes.
- If combination therapy with a thrombopoietic agent is used, any approved thrombopoietic agent at a standard starting dose may be used. The optimal duration of therapy with a thrombopoietic agent is uncertain; it is typically guided by clinical response, toxicity, and platelet count stability. If a thrombopoietic agent and corticosteroids are used together, in a patient who responds, it is advisable to taper the corticosteroids before the thrombopoietic agent. In some cases, corticosteroids may need to be tapered rapidly to avoid or minimize thrombocytosis.

Treatment in Adults with Primary ITP needing additional treatment after initial therapy with corticosteroids (with or without IVIG)

Recommendation 2a. In adults with primary ITP needing additional treatment after initial therapy with corticosteroids (with or without IVIG), the ASH guideline panel *recommends* a thrombopoietic agent (strong recommendation based on moderate certainty in the evidence of effects ⊕⊕⊕◯) or *suggests* rituximab (conditional recommendation based on low certainty in the evidence ⊕⊕◯◯).

Remarks:

- If a thrombopoietic agent is used, any approved thrombopoietic agent at a standard starting dose may be used. Duration of therapy is typically guided by clinical response, toxicity, and platelet count stability.
- Rituximab may be preferable for patients who value avoiding chronic medication, or who have a contraindication to, are unable to access, or cannot tolerate thrombopoietic agents.
- If rituximab is used, the most studied regimen is 375 mg/m2 × 4 weekly doses. However, other doses and schedules of rituximab may be used according to local practice. Biosimilars to rituximab are suitable substitutes.

Recommendation 2b. For adults with primary ITP needing additional treatment after initial therapy with corticosteroids (with or without IVIG), who are either ineligible for or choose not to receive a thrombopoietic agent or rituximab, the ASH guideline panel *suggests* a BTK inhibitor, MMF, or a SYK inhibitor (conditional recommendation based on very low certainty in the evidence ⊕◯◯◯).

Remarks:

- The panel concluded that current evidence in the population of interest is too indirect to definitively rank BTK inhibitors, MMF, or SYK inhibitors. Side effect profiles, cost, availability, and shared decision-making may be considered in selecting an agent.
- The panel concluded that azathioprine has a less favorable benefit to harm ratio; therefore, it is not equivalent to the other suggested treatment options. Nevertheless, azathioprine may be useful in some settings due to its low cost, global availability, and safety profile in pregnancy.

Good practice statements

Splenectomy

> If feasible, splenectomy should be delayed for at least one year after diagnosis because of the potential for remission.

> The treating clinician should ensure that patients receive appropriate immunizations at least two weeks prior to splenectomy, when feasible, and counseling regarding antibiotic prophylaxis and lifelong risk of sepsis and thrombosis following splenectomy. The treating clinician should educate the patient on prompt recognition and management of fever with antibiotics. After splenectomy, patients should be followed at least annually for platelet count monitoring, counseling, and to ensure adherence to recommended immunization schedules.

Thrombopoietic Agent Switching

> Switching to a different thrombopoietic agent may be considered based on insufficient response, adverse effects, comorbidities, poor adherence, or patient preference.
>
> Shared decision-making with the patient, documentation of the rationale for switching, avoidance of treatment gaps, and close monitoring around the time of switching are good practices to ensure optimal outcomes and continuity of care.

See the "good practice statements” broader section for background on splenectomy and thrombopoietic agent switching.

Values and preferences

These recommendations place a relatively higher value on patient-important outcomes (e.g., bleeding, fatigue, health-related quality of life, thrombosis, infection, sustained response off treatment) and a relatively lower value on surrogate outcomes (e.g., platelet count).

Explanations and other considerations

For both multi-comparison questions, the panel treated all thrombopoietic agents (i.e., avatrombopag, eltrombopag, hetrombopag, romiplostim, recombinant human thrombopoietin) as interchangeable. Our systematic literature review included the search terms: anti-CD20, rituximab, and Rituxan. Among the identified articles that met our inclusion criteria, the only anti-CD20 agent that was identified was rituximab.

Introduction

Aim of these guidelines and specific objectives

The purpose of these guidelines is to provide a focused update of the 2019 ASH ITP guidelines, specifically with respect to management of adults with primary ITP[15,16]. Through improved provider and patient education of the available evidence and evidence-based recommendations, these guidelines aim to provide clinical decision support for shared decision-making that will result in improved outcomes for adults with ITP.

The target audience includes patients, hematologists, general practitioners, internists, general surgeons, other clinicians and decision-makers. Policy makers interested in these guidelines include those involved in developing local, national or international plans with the goal to foster best practice, reduce costs, and improve patient outcomes. This document may also serve as the basis for adaptation by local, regional or national guideline panels.

Description of the health problem

ITP is an acquired disorder characterized by autoimmune-mediated platelet clearance and impairment of thrombopoiesis, resulting in thrombocytopenia[5]. It may occur in isolation (primary ITP) or in association with a predisposing condition (secondary ITP)[17]. ITP may present at any age with incidence peaks in early childhood and the elderly. The estimated incidence of ITP in adults is 1 to 6 per 100,000[7-11]. Common clinical manifestations include bleeding, fatigue, and reduced health-related quality of life[12]. Bleeding symptoms are highly variable among patients and show limited correlation with platelet count.

Treatment options for ITP have expanded rapidly over recent years. Since the 2019 ASH ITP guidelines were developed, inhibitors of splenic tyrosine kinase and Bruton tyrosine kinase have joined older treatment modalities including corticosteroids, IVIG, anti-D, rituximab, thrombopoietic agents, splenectomy, mycophenolate mofetil (MMF), and a host of other immunosuppressive medications in the ITP armamentarium. Moreover, the pipeline is brimming with new classes of therapeutics, some of which are in late-stage clinical development, including B-cell activating factor receptor inhibitors, anti-

CD38 agents, neonatal Fc-receptor antagonists, chimeric antigen receptor T-cell therapy, and complement inhibitors[6,13].

The explosion in treatment options for ITP has ushered in an era of new opportunities in clinical practice. With the success of currently available treatments in achieving and maintaining hemostatic platelet counts in most patients, the goalposts are shifting. No longer satisfied with platelet count-based outcomes alone, patients and clinicians have begun to focus on more ambitious goals including improved health-related quality of life, sustained response off treatment, and even cure. How best to achieve these lofty goals remains unsettled.

The growing list of treatments also presents a challenge. Patients and clinicians have more options than ever from which to choose but must decide how to prioritize, sequence, and potentially combine therapies. Guidance to aid in this determination is lacking.

These guidelines endeavor to tackle some of these opportunities and challenges. Question 1 addresses initial therapy in adults with primary ITP with an eye toward amelioration of the long-term disease course and minimization of corticosteroid-associated toxicity. Question 2 addresses treatment of adults with primary ITP needing additional treatment after initial therapy with corticosteroids with the goal of providing ranked options for second-line therapy and potential for sustained response off therapy. Good practice statements on thrombopoietic agent switching and the role of splenectomy in contemporary practice are also included.

As a focused update of the 2019 ASH ITP guidelines[15,16], these guidelines only address specific aspects of management in adults (≥18 years) with primary ITP. They do not address other important facets of the disease including diagnosis, indications for treatment, ITP in pregnancy, and pediatric ITP (age <18 years). Table 1 lists recommendations from the 2019 ASH ITP guidelines and notes whether they remain current or have been superseded by recommendations in the 2026 guidelines.

Table 1. Recommendations from the 2019 ASH ITP guidelines and their status following publication of the 2026 ASH ITP guidelines.

2019 recommendation	**Status with respect to the 2026 ASH ITP guidelines**
1a. In adults with newly diagnosed ITP and a platelet count of <30× 10^9/L who are asymptomatic or have minor mucocutaneous bleeding, the American Society of Hematology (ASH) guideline panel suggests corticosteroids rather than management with observation (conditional recommendation based on very low	The threshold for initiating therapy has not been updated.

certainty in the evidence of effects ⊕◯◯◯). **Remark**: There may be a subset of patients within this group for whom observation might be appropriate. This should include consideration of the severity of thrombocytopenia, additional comorbidities, use of anticoagulant or antiplatelet medications, need for upcoming procedures, and age of the patient.	However, the choice of first-line therapy has been updated in recommendation 1 of the 2026 ASH ITP guidelines.
1b. In adults with newly diagnosed ITP and a platelet count of ≥30× 10^9/L who are asymptomatic or have minor mucocutaneous bleeding, the ASH guideline panel recommends against corticosteroids and in favor of management with observation (strong recommendation based on very low certainty in the evidence of effects ⊕◯◯◯). **Remark**: For patients with a platelet count at the lower end of this threshold, for those with additional comorbidities, anticoagulant or antiplatelet medications, or upcoming procedures, and for elderly patients (>60 years old), treatment with corticosteroids may be appropriate.	
2a. In adults with newly diagnosed ITP and a platelet count of <20×10^9/L who are asymptomatic or have minor mucocutaneous bleeding, the ASH guideline panel suggests admission to the hospital rather than management as an outpatient (conditional recommendation based on very low certainty in the evidence of effects ⊕◯◯◯). In adults with an established diagnosis of ITP and a platelet count of <20×10^9/L who are asymptomatic or have minor mucocutaneous bleeding, the ASH guideline panel suggests outpatient management rather than hospital admission (conditional recommendation based on very low certainty in the evidence ⊕◯◯◯). **Remark**: Patients who are refractory to treatment, those with social concerns, uncertainty about the diagnosis, significant comorbidities with risk of bleeding, and more significant mucosal bleeding may benefit from admission to the hospital. Patients not admitted to the hospital should receive education and expedited follow-up with a hematologist. The need for admission is also variable across the range of platelet counts represented here (0 to 20×10^9/L).	Not updated
2b. In adults with a platelet count of ≥20×10^9/L who are asymptomatic or have minor mucocutaneous bleeding, the ASH guideline panel suggests management as an outpatient rather than hospital admission (conditional recommendation based on very low certainty in the evidence of effects ⊕◯◯◯). **Remark**: Patients who are refractory to treatment, with social concerns, uncertainty about the diagnosis, significant comorbidities with risk of bleeding, and more significant mucosal bleeding may benefit from admission to the hospital. Patients not admitted to the hospital should receive education and expedited follow-up with a hematologist. The need for admission is also variable across the range of platelet counts represented here (20×10^9/L to 150×10^9/L).	
3. In adults with newly diagnosed ITP, the ASH guideline panel recommends against a prolonged course (>6 weeks including treatment and taper) of prednisone and in favor of a short course	The duration of prednisone has

(≤6 weeks) (strong recommendation based on very low certainty in the evidence of effects ⨁◯◯◯).	not been updated. However, the choice of first-line therapy has been updated in recommendation 1 of the 2026 ASH ITP guidelines.
4. In adults with newly diagnosed ITP, the ASH guideline panel suggests either prednisone (0.5-2.0 mg/kg per day) or dexamethasone (40 mg per day for 4 days) as the type of corticosteroid for initial therapy (conditional recommendation based on very low certainty in the evidence of effects ⨁◯◯◯). **Remark**: If a high value is placed on rapidity of platelet count response, an initial course of dexamethasone may be preferred over prednisone, given that dexamethasone showed increased desirable effects with regard to response at 7 days.	The choice of prednisone vs. dexamethasone has not been updated. However, the choice of first-line therapy has been updated in recommendation 1 of the 2026 ASH ITP guidelines.
5. In adults with newly diagnosed ITP, the ASH guideline panel suggests corticosteroids alone rather than rituximab and corticosteroids for initial therapy (conditional recommendation based on very low certainty in the evidence of effects ⨁◯◯◯). **Remark**: If high value is placed on the possibility for remission over concerns for potential side effects of rituximab, then an initial course of corticosteroids with rituximab may be preferred.	This recommendation is retired and replaced with recommendation 1 of the 2026 ASH ITP guidelines.
6. In adults with ITP for ≥3 months who are corticosteroid-dependent or unresponsive to corticosteroids and are going to be treated with a thrombopoietin receptor agonist (TPO-RA), the ASH guideline panel suggests either eltrombopag or romiplostim (conditional recommendation based on very low certainty in the evidence of effects ⨁◯◯◯). **Remark**: Individual patient preference may place a higher value on the use of a daily oral medication or weekly subcutaneous injections.	This recommendation is retired and replaced with recommendation 2 of the 2026 ASH ITP guidelines.
7. In adults with ITP lasting ≥3 months who are corticosteroid-dependent or have no response to corticosteroids, the ASH guideline panel suggests either splenectomy or a TPO-RA (conditional recommendation based on very low certainty in the evidence of effects ⨁◯◯◯).	This recommendation is retired and replaced with recommendation 2 of the 2026 ASH ITP guidelines.
8. In adults with ITP lasting ≥3 months who are corticosteroid-dependent or have no response to corticosteroids, the ASH guideline panel suggests rituximab	This recommendation is retired and

rather than splenectomy (conditional recommendation based on very low certainty in the evidence of effects ⨁◯◯◯).	replaced with recommendation 2 of the 2026 ASH ITP guidelines.
9. In adults with ITP lasting ≥3 months who are corticosteroid-dependent or have no response to corticosteroids, the ASH guideline panel suggests a TPO-RA rather than rituximab (conditional recommendation based on very low certainty in the evidence of effects ⨁◯◯◯). **Remark**: These recommendations are the result of dichotomous evaluation of treatments that are often being considered simultaneously. Each of these second line treatments may be effective therapy and therefore the choice of treatment should be individualized based on duration of ITP, frequency of bleeding episodes requiring hospitalization or rescue medication, comorbidities, age of the patient, medication adherence, medical and social support networks, patient values and preferences, cost, and availability. Patient education and shared decision-making are encouraged. If possible, splenectomy should be delayed for at least 1 year after diagnosis because of the potential for spontaneous remission in the first year. Patients who value avoidance of long-term medication may prefer splenectomy or rituximab. Patients who wish to avoid surgery may prefer a TPO-RA or rituximab. Patients who place a high value on achieving a durable response may prefer splenectomy or TPO-RAs.	This recommendation is retired and replaced with recommendation 2 of the 2026 ASH ITP guidelines.
10a. In children with newly diagnosed ITP and a platelet count of <20 $\times 10^9$/L who have no or mild bleeding (skin manifestations) only, the ASH guideline panel suggests against admission to the hospital and in favor of management as an outpatient (conditional recommendation based on very low certainty in the evidence of effects ⨁◯◯◯). **Remark**: For patients with uncertainty about the diagnosis, those with social concerns, those who live far from the hospital, and those for whom follow-up cannot be guaranteed, admission to the hospital may be preferable.	Not updated
10b. In children with newly diagnosed ITP and a platelet count of ≥20×10^9/L who have no or mild bleeding (skin manifestations) only, the ASH guideline panel suggests against admission to the hospital and in favor of management as an outpatient (conditional recommendation based on very low certainty in the evidence of effects ⨁◯◯◯). **Remark**: For patients with uncertainty about the diagnosis, those with social concerns, those who live far from the hospital, or those for whom followup cannot be guaranteed, admission to the hospital may be preferable.	
11. In children with newly diagnosed ITP who have no or minor bleeding, the ASH guideline panel suggests observation rather than corticosteroids (conditional recommendation based on very low certainty in the evidence of effects ⨁◯◯◯).	Not updated
12. In children with newly diagnosed ITP who have no or minor bleeding, the ASH guideline panel recommends observation rather than IV immunoglobulin	Not updated

(IVIG) (strong recommendation based on moderate certainty in the evidence of effects ⨁⨁⨁◯).	
13. In children with newly diagnosed ITP who have no or minor bleeding, the ASH guideline panel recommends observation rather than anti-D immunoglobulin (strong recommendation based on moderate certainty in the evidence of effects ⨁⨁⨁◯).	Not updated
14. In children with newly diagnosed ITP who have non–life-threatening mucosal bleeding and/or diminished HRQoL, the ASH guideline panel recommends against courses of corticosteroids longer than 7 days and in favor of courses 7 days or shorter (strong recommendation based on very low certainty in the evidence of effects ⨁◯◯◯).	Not updated
15. In children with newly diagnosed ITP who have non–life-threatening mucosal bleeding and/or diminished HRQoL, the ASH guideline panel suggests prednisone (2-4 mg/kg per day; maximum, 120 mg daily, for 5-7 days) rather than dexamethasone (0.6 mg/kg per day; maximum, 40 mg per day for 4 days) (conditional recommendation based on very low certainty in the evidence of effects ⨁◯◯◯).	Not updated
16. In children with newly diagnosed ITP who have non–life-threatening mucosal bleeding and/or diminished HRQoL, the ASH guideline panel suggests corticosteroids rather than anti-D immunoglobulin (conditional recommendation based on low certainty in the evidence of effects ⨁⨁◯◯). **Remark**: This recommendation assumes corticosteroid dosing as outlined in recommendations 14 and 15. This recommendation is reserved only for children with nonmajor mucosal bleeding.	Not updated
17. In children with newly diagnosed ITP who have non–life-threatening mucosal bleeding and/or diminished HRQoL, the ASH guideline panel suggests either anti-D immunoglobulin or IVIG (conditional recommendation based on low certainty in the evidence of effects ⨁⨁◯◯). **Remark**: This recommendation is reserved only for children with nonmajor mucosal bleeding.	Not updated
18. In children with newly diagnosed ITP who have non–life-threatening mucosal bleeding and/or diminished HRQoL, the ASH guideline panel suggests corticosteroids rather than IVIG (conditional recommendation based on low certainty in the evidence of effects ⨁⨁◯◯). **Remark**: This recommendation assumes that a short course of corticosteroids is being used for treatment as recommended in recommendation 14. This recommendation is reserved only for children with nonmajor mucosal bleeding.	Not updated
19. In children with ITP who have non–life-threatening mucosal bleeding and/or diminished HRQoL and do not respond to first-line treatment, the ASH guideline panel suggests the use of TPO-RAs rather than rituximab (conditional recommendation based on very low certainty in the evidence of effects ⨁◯◯◯).	Not updated
20. In children with ITP who have non–life-threatening mucosal bleeding and/or diminished HRQoL and do not respond to first-line treatment, the ASH guideline panel suggests TPO-RAs rather than splenectomy (conditional recommendation based on very low certainty in the evidence of effects ⨁◯◯◯).	Not updated

21. In children with ITP who have non–life-threatening mucosal bleeding and/or diminished HRQoL and do not respond to first-line treatment, the ASH guideline panel suggests rituximab rather than splenectomy (conditional recommendation based on very low certainty in the evidence of effects ⨁◯◯◯).	Not updated

Methods

The guideline panel developed and graded the recommendations and assessed the certainty of the supporting evidence following the GRADE approach[4]. ASH guided the overall guideline development process, including project funding, panel formation, conflict of interest management, internal and external review, and organizational approval, based on policies and procedures intended to meet recommendations for trustworthy guidelines by the Institute of Medicine and the Guidelines International Network[1-3].

In ITP, there are multi ple interventions that could simultaneously be considered for second-line therapy. To address this, the guideline panel utilized the GRADEpro multiple comparison feature of the GRADEpro software as part of the Evidence to Decision (EtD) frameworks[18].

Organization, panel composition, planning and coordination

The work of this panel was coordinated by ASH and the University of Oklahoma (funded by ASH under a paid agreement). Project oversight was provided by the ASH Guideline Oversight Subcommittee, which reported to the ASH Committee on Quality. ASH vetted and appointed individuals to the guideline panel. The University of Oklahoma conducted systematic reviews of evidence and coordinated the guideline development process including the use of the GRADE approach. The membership of the panel and the systematic review team is described in **Supplement 1**.

The panel included adult and pediatric hematologist-oncologists, clinical and methodological experts, and both patient and caregiver representatives, all with expertise in ITP. One co-chair was a clinical content expert; the other co-chair was an expert in guideline development methodology. To promote efficiency and continuity with prior work, approximately half of the guideline panel, including the clinical co-chair, was drawn from the panel that authored the 2019 guidelines, while the remaining members were newly appointed.

In addition to synthesizing evidence systematically, the University of Oklahoma supported the guideline development process, including determining methods, preparing meeting materials, and facilitating panel discussions. The panel's work was done using web-based tools (www.gradepro.org) and face-to-face and online meetings via Zoom.

Guideline funding and management of conflicts of interest

Development of these guidelines was wholly funded by ASH, a nonprofit medical specialty society that represents hematologists. Most members of the guideline panel were members of ASH. ASH staff supported panel appointments and coordinated meetings but had no role in choosing the guideline questions or determining the recommendations.

Members of the guideline panel received travel reimbursement for attendance at in-person meetings. Through the University of Oklahoma, some researchers who contributed to the systematic evidence reviews received salary support. Other researchers participated to fulfill requirements of an academic degree or program.

Conflicts of interest of all participants were managed according to ASH policies based on recommendations of the Institute of Medicine[1] and the Guidelines International Network[3]. Participants disclosed all financial and nonfinancial interests, including research funding, consulting, speakers' bureaus, and other relationships. ASH staff and appointed experts from ASH membership reviewed the disclosures and proposed the guideline panel to include a diversity of expertise and perspectives and avoid panel members having relevant conflicts. The proposed panel was reviewed and approved by the ASH Committee on Quality and Executive Committee. During the guideline development process, all panelists avoided significant direct financial relationships with for-profit health care companies. Many panelists had relevant nonfinancial interests, such as published opinions on the guideline topic. None of the researchers who contributed to the systematic evidence reviews or who supported the guideline development process had any such conflicts.

Recusal procedures were established *a priori* to manage certain conflicts[4]. During deliberations about recommendations, any panel member with an active principal investigator role with a for-profit healthcare company that could be affected by a recommendation based on the prioritized questions (i.e., a commercial entity that marketed any product that could be affected by that recommendation) would have participated in discussions about the evidence and clinical context but would have been recused from making judgments or voting about the direction and strength of the recommendation. In this guideline, no panelist met these recusal criteria, and therefore no recusals occurred.

Supplement 2 provides the complete participant information forms of all panel members, including disclosures prior and during participation in the panel. In part A of the forms, individuals disclosed direct financial interests for two years prior to appointment; in part B, indirect financial interests; and in part C, non-financial interests. Part D describes new interests disclosed by individuals after appointment. Part E summarizes ASH decisions about which interests were judged to be conflicts and how they were managed through disclosure.

Supplement 3 provides the complete participant information forms of researchers who contributed to these guidelines.

Formulating specific clinical questions and determining outcomes of interest

The panel used the GRADEpro Guideline Development Tool (www.gradepro.org)[19] to brainstorm and then prioritize the questions described in **Table 2**. The Multiple Comparison feature in GRADEpro allowed the guideline panel to make judgements and comparisons across several interventions simultaneously. Initially, the panel members compared each intervention to the standard of care. After judgements were made for each intervention, then comparisons were made across the multiple interventions. Unlike pairwise comparisons, the GRADEpro multiple comparison feature extends the standard EtD framework to multiple comparisons, allowing the guideline panel to assess not only treatment effects, but also benefits, harms, certainty of evidence, values, resources required, equity, acceptability and feasibility.

The panel selected outcomes of interest for each question *a priori*, following the approach described in detail elsewhere[4]. In brief, the panel first brainstormed all possible outcomes before rating their relative importance for decision-making following the GRADE approach[4]. While acknowledging considerable variation in the impact on patient outcomes, the panel considered the following outcomes in Table 2 as critical for clinical decision-making across the two questions: sustained response off treatment (SRoT) at 6 months, SRoT at 12 months, health-related quality of life (HRQoL), major/minor bleeding, thrombosis, infection, diarrhea/gastrointestinal symptoms, durable response at 6 months, durable response at 12 months, time to response, and reduction in need for rescue therapy. For PICO 1, we also considered length of corticosteroid treatment. We also considered individual domains of HRQoL that are relevant to ITP such as fatigue, anxiety and depression.

Table 2. Questions and outcomes prioritized by the guideline panel

	PICO Question	Outcomes
1	Should adults with primary ITP requiring initial therapy, be treated with corticosteroids (with or without IVIG) + additional agent (MMF, thrombopoietic agent, anti-CD20, SYK inhibitor) or corticosteroids (with or without IVIG)?	• Length of corticosteroid treatment-steroid exposure • SRoT at 6 months • SRoT at 12 months • Health related quality of life • Major/minor bleeding • Thrombosis • Infection • Diarrhea/GI symptoms
2	In adults with primary ITP needing additional therapy after initial treatment with corticosteroids (with or without IVIG), should thrombopoietic agents, or anti-CD20, or SYK inhibitors, or BTK inhibitors (rilzabrutinib), or MMF, or anti-CD38 or azathioprine be used?	• Durable response (platelet count >30 × 10^9/L and doubling from baseline at 6 months, on or off treatment) • Time to response • SRoT at 6 months • SRoT at 12 months • Reduce the need for rescue therapy • Health related quality of life • Major/minor bleeding • Thrombosis • Increased risk of infection • Diarrhea/GI symptoms

Evidence review and development of recommendations

For each guideline question, the University of Oklahoma prepared a GRADE "Evidence-to-Decision" (EtD) framework, using the GRADEpro Guideline Development Tool (www.gradepro.org) [20-22]. The EtD table summarized the results of systematic reviews of the literature that were updated or performed for this guideline. The panel met twice monthly to review the results of the systematic reviews and to confirm inclusion/exclusion of the identified studies. The EtD table addressed effects of interventions, resource utilization (cost-effectiveness), values and preferences (relative importance of outcomes), equity, acceptability and feasibility. The guideline panel reviewed the draft EtD tables during and after the guideline panel meeting and made suggestions for corrections. To ensure that recent studies were not missed, searches (presented in **Supplement 4**) were updated July 19, 2025, and panel members were asked to suggest any studies that may have been missed and fulfilled the inclusion criteria for the individual questions.

Under the direction of the University of Oklahoma, researchers followed the general methods outlined in the Cochrane Handbook for Systematic Reviews of Interventions (handbook.cochrane.org) for conducting updated or new systematic reviews of intervention effects. Detailed search strategies, data collection procedures, risk-of-bias assessments, evidence synthesis methods, and certainty-of-evidence ratings are described in **Supplement 5.** Included studies ranged in publication dates and eligibility criteria. Although the included studies had different primary endpoints, data were abstracted for any reported endpoint of interest regardless of whether it was the primary endpoint of the study. When existing reviews were used, judgments of the original authors about risk of bias were either randomly checked for accuracy and accepted or conducted *de novo* if they were not available or not reproducible.For new reviews, risk of bias was assessed at the health outcome level using the Cochrane Collaboration's risk of bias tool for randomized trials or non-randomized studies[23,24]. In addition to conducting systematic reviews of intervention effects, the researchers searched for evidence related to baseline risks, values, preferences and costs, and summarized findings within the EtD frameworks[20-22]. Subsequently, the certainty of the body of evidence (also known as quality of the evidence or confidence in the estimated effects) was assessed for each effect estimate of the outcomes of interest following the GRADE approach based on the following domains: risk of bias, precision, consistency and magnitude of the estimates of effects, directness of the evidence, risk of publication bias, presence of large effects, dose-response relationship, and an assessment of the effect of residual, opposing confounding. The certainty was categorized into four levels ranging from very low to high[44]. Within this report, these categories are represented by symbols, as described in Figure 2. Interested readers may find more explanation about the GRADE approach to assessing and rating certainty in a body of evidence in other publications[4].

The original PICO 1 question included SYK inhibitors. A systematic literature review was conducted, and no studies were identified that met our inclusion criteria. Therefore, SYK inhibitors were removed from the PICO 1 question. The original PICO 2 question included anti-CD38 agents. A systematic literature review was conducted, and one study was identified that met our inclusion criteria. It was small (N=22) and did not include a control group. Therefore, the panel determined there was insufficient evidence to make a recommendation on anti-CD38 agents. PICO 2 assumes that patients were not treated upfront with rituximab or a thrombopoietic agent as recommended in PICO 1. For more information on the identification of studies, screening, and inclusion for both PICOs 1 and 2, please see **Supplement 6**.

During a 2-day in-person meeting followed by online communication and conference calls, the panel developed clinical recommendations based on the evidence summarized in the EtD tables. For each recommendation, the panel came to consensus on the following: the certainty of the evidence, the balance of benefits and harms of the compared management options and the assumptions about the values and preferences associated with the decision. The guideline panel also explicitly took into account the extent of resource use associated with alternative management options. The panel agreed on the recommendations (including direction and strength), remarks, and qualifications by consensus or, if necessary, by voting (an 80% majority was required for a strong recommendation), based on the balance of all desirable and undesirable consequences. The final guidelines, including recommendations, were reviewed and approved by all members of the panel.

Interpretation of strong and conditional recommendations

The recommendations are labeled as either “strong” or “conditional” according to the GRADE approach. The words “the guideline panel recommends” are used for strong recommendations, and “the guideline panel suggests” for conditional recommendations. Figure 1 provides GRADE’s interpretation of strong and conditional recommendations by patients, clinicians, health care policy makers and researchers.

Document review

Draft recommendations were reviewed by all members of the panel, revised, and then made available online on December 12, 2025, for external review by stakeholders including allied organizations, other medical professionals, patients, and the public. Sixteen individuals or organizations submitted comments. Remarks were revised to address pertinent feedback; however, no changes were made to the recommendations themselves. On May 21, 2026, the ASH Guideline Oversight Subcommittee and the ASH Committee on Quality approved that the defined guideline development process was followed, and on May 26, 2026, the officers of the ASH Executive Committee approved submission of the guidelines for publication under the imprimatur of ASH. The guidelines were then subjected to peer review by *Blood Advances.*

How to use these guidelines

ASH guidelines are primarily intended to help clinicians and patients make shared decisions about diagnostic and treatment alternatives, with the goal of improving quality of patient care. Other purposes are to inform policy, education and advocacy, and to state future research needs. These guidelines are not intended to serve or be construed as a standard of care. Clinicians must make decisions on the basis of the clinical presentation of each individual patient, ideally through a shared process that considers the patient's values and preferences with respect to the anticipated outcomes of the chosen option.

Decisions may be constrained by the realities of a specific clinical setting and local resources, including but not limited to institutional policies, time limitations, or availability of treatments. These guidelines may not include all appropriate methods of care for the clinical scenarios described. As science advances and new evidence becomes available, recommendations may become outdated. Following these guidelines cannot guarantee successful outcomes. ASH does not warrant or guarantee any products described in these guidelines.

Statements about the underlying values and preferences as well as qualifying remarks accompanying each recommendation are its integral parts and serve to facilitate more accurate interpretation. They should never be omitted when quoting or translating recommendations from these guidelines. The use of these guidelines is also facilitated by the links to the EtD frameworks and interactive summary of findings tables in each section.

Recommendations

The recommendations below represent a focused update of the 2019 ASH ITP guidelines (see Table 1 for the recommendations covered in this document)[15]. Recommendation 1 addresses initial therapy and recommendation 2 addresses management after failure of first-line corticosteroids in adults (≥18 years) with primary ITP (see Figure 3).

Initial Therapy in Adults with Primary ITP

In adults with primary ITP requiring initial therapy should corticosteroids (with or without IVIG) plus an additional agent (anti-CD20, MMF, thrombopoietic agent), or corticosteroids (with or without IVIG) alone be used?

Recommendation 1. In adults with primary ITP requiring initial therapy, the ASH guideline panel *suggests either* rituximab plus corticosteroids (with or without IVIG) or a thrombopoietic agent plus corticosteroids (with or without IVIG) rather than mycophenolate mofetil plus corticosteroids (with or without IVIG) or corticosteroids (with or without IVIG) alone [conditional recommendation based on low certainty in the evidence ⊕⊕◯◯ (recommendation limited to rituximab or its biosimilars)].

If rituximab and thrombopoietic agents are not available options, the panel *suggests* corticosteroids (with or without IVIG) alone rather than mycophenolate mofetil (MMF) plus corticosteroids (with or without IVIG) [conditional recommendation based on very low certainty in the evidence ⊕◯◯◯].

Remarks:

- A demonstrable rise in platelet count with corticosteroids and/or IVIG increases confidence in the diagnosis of ITP. Therefore, it is reasonable to treat a patient with a brief course (≤2 weeks) of corticosteroids (with or without IVIG) to increase confidence in the diagnosis of ITP before implementing combination therapy, especially in cases of diagnostic uncertainty.
- Although the most studied corticosteroid component of combination therapy is dexamethasone 40 mg daily × 4 days for 1 to 3 cycles spaced 2 to 4 weeks apart, it is also appropriate to use prednis(ol)one at a starting dose of 0.5-2 mg/kg/day.
- Whether corticosteroids are used as part of combination therapy or alone, it is advisable to limit their administration to 6 weeks or less to minimize toxicity.

- If combination therapy with rituximab is used, the most studied regimen is 375 mg/m2 × 4 weekly doses. However, other doses and schedules of rituximab may be used according to local practice. Biosimilars to rituximab are suitable substitutes.
- If combination therapy with a thrombopoietic agent is used, any approved thrombopoietic agent at a standard starting dose may be used. The optimal duration of therapy with a thrombopoietic agent is uncertain; it is typically guided by clinical response, toxicity, and platelet count stability. If a thrombopoietic agent and corticosteroids are used together, in a patient who responds, it is advisable to taper the corticosteroids before the thrombopoietic agent. In some cases, corticosteroids may need to be tapered rapidly to avoid or minimize thrombocytosis.

Summary of the evidence

This multi-comparison question compared three different treatments (MMF, thrombopoietic agent, anti-CD20) plus corticosteroids with or without IVIG to a uniform standard of care (SOC) which was defined as corticosteroids with or without IVIG. **Supplement 7a** presents the characteristics of all included studies for PICO 1 for MMF, thrombopoietic agents, and anti-CD20. **Supplement 7b** presents study level data for thrombopoietic and anti-CD20 agents for SRoT and HRQoL outcomes.

Mycophenolate Mofetil (MMF)

We found 1 randomized clinical trial (RCT)[25] that addressed this question. The study included newly diagnosed treatment-naïve adults with ITP.

The RCT reported the effect of MMF on durable response at 12 months, major and minor bleeding, infection, overall gastrointestinal (GI) symptoms, time to response, need for rescue medications and HRQoL [25].

The evidence profile and EtD framework for this intervention is available online at https://guidelines.ash.gradepro.org/profile/pB_qrthRaVU.

Thrombopoietic agents

We found 7 studies, 3 RCTs[26-28] and 4 prospective one-armed cohort studies [29-32], that addressed this question. Six studies included newly diagnosed treatment-naïve adults with ITP and were classified as direct evidence [26,27,29-32] and one study included treatment-naïve adults with ITP that could have newly diagnosed, persistent or chronic ITP and was therefore classified as indirect evidence [28]. The data from studies without a comparator arm were reported in the evidence tables, assessed by the panel, and informed the recommendations, but are only reported in the text when there is no evidence from the RCTs.

Four studies reported sustained response off treatment (SRoT) at 6 months [26,27,29,30], 3 studies reported SRoT at 12 months [27,29,30], 3 studies assessed durable response at 6 months [28,29,31], 1 study assessed durable response at 12 months [29], 1 study assessed major bleeding[27], 4 studies assessed minor bleeding[26,29-31], 4 studies assessed infection[26-28,30], 1 study assessed overall GI symptoms[27], five studies assessed thrombosis[26,27,29,30,32], 1 study assessed nausea[31], 1 study assessed vomiting[31], 1 study reported anxiety[26], 1 study reported depression[26], 1 study reported need for rescue therapy[26], and 4 studies reported fatigue[26,27,30,31].

The evidence profile and EtD framework for this intervention is available online at https://guidelines.ash.gradepro.org/profile/EautU3Ze0ow.

Anti-CD20

We found 6 studies, 3 RCTs [26,33,34], 1 case-control study[35], and 2 retrospective one-armed cohort studies[36,37] that addressed this question. Five studies included newly diagnosed treatment-naïve adults with ITP and were classified as direct evidence [26,33,35-37] and 1 study included treatment-naïve adults with ITP that could have newly diagnosed, persistent or chronic ITP and was therefore classified as indirect evidence[34]. The data from studies without a comparator arm were reported in the evidence tables, assessed by the panel, and informed the recommendations, but are only reported in the text when there is no evidence from the controlled studies.

Five studies reported sustained response off treatment (SRoT) at 6 months[26,33-36], 4 studies reported SRoT at 12 months[33,35-37], 1 study assessed durable response at 12 months[37], 1 study assessed major bleeding[33], 1 study assessed minor bleeding[26], 4 studies assessed infection[26,33-35], 1 study assessed overall GI symptoms[33], 2 studies assessed thrombosis[26,34], 1 study reported anxiety[26], 1 study reported depression[26], 1 study reported need for rescue therapy[26], 1 study reported fatigue[26,27,30,31], and 1 study reported time to response[36].

The evidence profile and EtD framework for this intervention is available online at https://guidelines.ash.gradepro.org/profile/JNgz58cv_9U.

Benefits

MMF

MMF increased the odds of durable response at 12 months compared to the SOC (46 of 59 (78%) MMF versus 34 of 61 (56%) SOC; odds ratio (OR): 2.81, 95% confidence interval (CI): 1.27 to 6.23). MMF also may reduce the odds for need for rescue therapy (8 of 59 (14%) MMF versus 10 of 61 (16%) SOC; OR: 0.80, 95% CI: 0.29 to 2.19) and improved time to response (median 4 days shorter for MMF; no 95% CI calculated) but the estimates were imprecise and the CI included no effect. The study assessing minor bleeding, infection, and major bleeding found no difference between those receiving MMF and SOC (minor bleeding: 14 of 59 (24%) MMF versus 15 of 61 (25%) SOC, OR: 0.95, 95% CI: 0.41, 2.20; infection: 14 of 59 (24%) MMF versus 14 of 61 (23%) SOC, OR: 1.04, 95% CI: 0.45, 2.43; major bleeding: 1 of 59 (2%) MMF versus 0 of 61 (0%) SOC, risk difference [RD]: 0.02, 95% CI: -0.02 to 0.05). Overall, the certainty of these estimated effects is very low owing to serious inconsistency and imprecision of the estimates due to the inclusion of only one moderate sized RCT. The panel deemed the benefits to be small to moderate but downgraded to small since SRoT, a prioritized outcome, was not available and durable response had to be used as a surrogate outcome.

Thrombopoietic agents

Thrombopoietic agents increased the odds of SRoT at 6 and 12 months (direct: 6 months: 136 of 260 (52%) thrombopoietic agent versus 58 of 175 (33%) SOC, OR: 2.33, 95% CI: 1.48 to 3.37; 12 months: 46 of 100 (46%) thrombopoietic agent versus 30 of 96 (31%) SOC, OR: 1.87, 95% CI: 1.05 to 3.36) and durable response at 6 months (indirect: 22 of 25 (88%) thrombopoietic agent versus 30 of 50 (60%) SOC, OR: 4.89, 95% CI: 1.29 to 18.53)) compared to SOC. Thrombopoietic agents decreased the odds of anxiety (direct: 104 of 188 (55%) thrombopoietic agent versus 94 of 105 (90%) SOC, OR: 0.14, 95% CI: 0.07 to 0.29), depression (direct: 86 of 188 (46%) thrombopoietic agent versus 99 of 105 (94%) SOC, OR: 0.05, 95% CI: 0.02 to 0.12) and minor bleeding (direct: 29 of 188 (15%) thrombopoietic agent versus 62 of 105 (59%) SOC, OR: 0.13, 95% CI: 0.07 to 0.22) compared to SOC. Thrombopoietic agents also may reduce the odds of fatigue (direct: 71 of 290 (25%) thrombopoietic agent versus 88 of 201 (44%) SOC, OR: 0.40, 95% CI: 0.05 to 3.27), need for rescue therapy (direct: 16 of 188 (9%) thrombopoietic agent versus 15 of 105 (14%) SOC, OR: 0.56, 95% CI: 0.26 to 1.18), and infection (direct: 12 of 290 (4%) thrombopoietic agent versus 26 of 201 (13%) SOC, OR: 0.20, 95% CI: 0.00 to 10.39; indirect: 0 of 25 (0%) thrombopoietic agent versus 4 of 50 (8%) SOC, OR: 0.20, 95% CI: 0.01 to 3.92) but the estimates are imprecise and CIs include no effect. Studies assessing overall gastrointestinal symptoms (direct: 9 of 102 (9%) thrombopoietic agent versus 7 of 96 (7%) SOC, OR: 1.23, 95% CI: 0.44 to 3.45), found no difference between those receiving a thrombopoietic agent and SOC groups. Overall, the certainty of these estimated effects is low due to risk of bias, inconsistency and imprecision of the estimates. The panel judged the desirable effects of thrombopoietic agents to be moderate, mainly driven by increased SRoT at 6 and 12 months, reduction in minor bleeding, reduction in need for rescue therapy, and improvement in various HRQoL dimensions. The panel noted that duration of treatment with a thrombopoietic agent was variable as was thrombopoietic agent dosing among included studies.

Anti-CD20

Anti-CD20 agents may increase the odds of SRoT at 6 months: (direct: 63 of 122 (52%) anti-CD20 versus 52 of 162 (32%) SOC, OR: 2.33, 95% CI: 0.88 to 6.12; indirect: 31 of 49 (63%) anti-CD20 versus 19 of 52 (37%) SOC, OR: 2.99, 95% CI: 1.33 to 6.72), and SRoT 12 months (direct: 47 of 80 (59%) anti-CD20 versus 25 of 81 (31%) SOC, OR: 4.23, 95% CI: 0.98 to 18.32). Anti-CD20 decreased the odds of minor bleeding (9 of 64 (14%) anti-CD20 versus 62 of 105 (59%) SOC, OR: 0.11, 95% CI: 0.05 to 0.25), anxiety (5 of 64 (8%) anti-CD20 versus 94 of 105 (90%) SOC, OR: 0.01, 95% CI: 0.00 to 0.03), depression (11 of 64 (17%) anti-CD20 versus 99 of 105 (94%) SOC, OR: 0.01, 95% CI: 0.00 to 0.04), fatigue (2 of 64 (3%) anti-CD20 versus 88 of 105 (77%) SOC, OR: 0.01, 95% CI: 0.00 to 0.04), and need for rescue therapy (1 of 64 (2%) anti-CD20 versus 15 of 105 (14%) SOC, OR: 0.10, 95% CI: 0.01 to 0.74). Overall, the certainty of these estimated effects is low due to some outcomes with serious inconsistency and serious imprecision in the estimates related to number of studies and heterogeneity of study designs and due to the indirectness of one study. The panel judged the desirable effects of anti-CD20 agents to be moderate, driven primarily by increased SRoT at 6 and 12 months, reduction in minor bleeding, reduction in need for rescue therapy, and improvement in various HRQoL dimensions. The panel noted that the effect of anti-CD20 on B-cell depletion and memory of B cells can last for many months; thus ‘exposure’ to the pharmacodynamic effects of the drug persists well after treatment is completed.

Harms and burden

MMF

One study reported adverse effects [25] with a decrease in a domain of HRQoL as measured by the FACT-F (fatigue) with a median score 3.3 points lower (95% CI: 0.04 lower to 6.6 points lower) in the MMF group as compared to the SOC group. MMF also may increase the odds for overall gastrointestinal (GI) symptoms (20 of 59 (34%) MMF versus 15 of 61 (25%) SOC, OR: 1.57, 95% CI: 0.71 to 3.48), but the estimate is imprecise and the confidence interval includes no effect. There is very low certainty in the estimate of the risk of adverse effects due to serious inconsistency and imprecision due to the inclusion of only one moderate sized RCT. However, given the available evidence, the guideline panel considered the undesirable effects to be moderate, mainly driven by negative impact on HRQoL.

Thrombopoietic agents

Thrombopoietic agents may increase the odds of major bleeding (direct: 2 of 102 (2%) thrombopoietic agent versus 0 of 96 (0%) SOC, OR: 4.80, 95% CI: 0.23 to 101.29) and thrombosis (direct: 29 of 290 (10%) thrombopoietic agent versus 0 of 201 (0%) SOC; OR: 11.49, 95% CI: 0.76 to 174.32), but the estimates are imprecise and the CIs include no effect. There is low certainty in the estimate of the risk of adverse effects due to risk of bias, inconsistency and imprecision of the estimates. However, given the available evidence, the guideline panel considered the undesirable effects to be small, mostly driven by a potential increased risk of thrombosis. The panel noted that some studies excluded patients with risk factors for thrombosis and that patients with prothrombotic risk factors are likely to be at greater risk.

Anti-CD20

Anti-CD20 agents increased the odds of thrombosis (9 of 64 (14%) anti-CD20 versus 0 of 105 (0%) SOC, OR: 36.12, 95% CI: 2.06 to 632.13) and infection (direct: 38 of 147 (26%) anti-CD20 versus 26 of 196 (13%) SOC, OR: 2.50, 95% CI: 1.01 to 6.21; indirect: 1 of 49 (2%) anti-CD20 versus 0 of 52 (0%) SOC, OR: 3.25, 95% CI: 0.13 to 81.63). Anti-CD20 may increase the odds of major bleeding (2 of 62 (3%) anti-CD20 versus 1 of 71 (1%) SOC, OR: 2.33, 95% CI: 0.21 to 26.37) and GI symptoms (14 of 62 (23%) anti-CD20 versus 11 of 71 (16%) SOC, OR: 1.59, 95% CI: 0.66 to 3.83, but the estimates are imprecise and the CIs include no effect. There is low certainty in the estimate of the risk of adverse effects due to some outcomes with serious inconsistency, serious imprecision, and serious indirectness of the estimates related to number of studies and heterogeneity of study designs. However, given the available evidence, the guideline panel considered the undesirable effects for anti-CD20 to be small, mainly driven by increased risk of infection. The panel noted that serious infections (e.g., progressive multifocal leukoencephalopathy) are very rare in patients with ITP. The panel noted that treatment with anti-CD20 attenuates the response to vaccinations for at least 6 months. The panel noted other important adverse effects that were not prioritized such as hepatitis B reactivation, neutropenia, serum sickness, and infusion reactions.

Other EtD criteria and considerations

The guideline panel noted that the cost of MMF is negligible whereas the cost of rituximab and thrombopoietic agents is moderate. The panel further determined that rituximab and thrombopoietic agents would likely reduce equity due to cost and uneven insurance coverage. The panel judged that MMF, rituximab, and thrombopoietic agents are probably acceptable and feasible, with cost and insurance coverage being the most significant barriers to their use as initial therapy. The panel underscored that MMF is contraindicated in pregnancy and that the safety of rituximab and the thrombopoietic agents is not well-established in pregnancy.

Conclusions for this recommendation

Results of the panel discussion of the pairwise comparisons informed the multi-comparison EtD (linked with each intervention) that was used to form final recommendations. The panel made a conditional recommendation in favor of combination therapy with rituximab plus corticosteroids (with or without IVIG) or a thrombopoietic agent plus corticosteroids (with or without IVIG) rather than corticosteroids (with or without IVIG) alone, primarily because the combination regimens demonstrated greater SRoT at 6 and 12 months, reduced minor bleeding, reduced rescue therapy, and improvement in various dimensions of HRQoL. These benefits were felt to more than offset the greater costs and reduced equity associated with the combination regimens.

An individualized approach to selection of initial therapy for adults with primary ITP corresponding to Recommendation 1 and incorporating patient values, preferences, and treatment availability is recommended and is summarized in Figure 3.

MMF plus corticosteroids (with or without IVIG) was rated less favorably by the guideline panel because there was no evidence regarding the prioritized outcome of SRoT with this regimen. Furthermore, HRQoL was significantly lower in the combination group than the corticosteroid-only group despite a higher rate of platelet count responses in the former.

In the event that neither rituximab nor thrombopoietic agents are available, the panel made a conditional recommendation in favor of initial therapy with corticosteroids (with or without IVIG) alone rather than combination therapy with MMF plus corticosteroids (with or without IVIG). This recommendation was primarily driven by decreased HRQoL, increased costs, and a lack of data on SRoT associated with the combination regimen.

Treatment in Adults with Primary ITP needing additional treatment after initial therapy with corticosteroids (with or without IVIG)

In adults with primary ITP needing additional treatment after initial therapy with corticosteroids (with or without IVIG), should an anti-CD20 agent, azathioprine, a BTK inhibitor, MMF, a SYK inhibitor, or a thrombopoietic agent be used, or should corticosteroids (with or without IVIG) be continued?

Recommendation 2a. In adults with primary ITP needing additional treatment after initial therapy with corticosteroids (with or without IVIG), the ASH guideline panel *recommends* a thrombopoietic agent (strong recommendation based on moderate certainty in the evidence of effects ⊕⊕⊕◯) or *suggests* rituximab (conditional recommendation based on low certainty in the evidence ⊕⊕◯◯).

Remarks:

- If a thrombopoietic agent is used, any approved thrombopoietic agent at a standard starting dose may be used. Duration of therapy is typically guided by clinical response, toxicity, and platelet count stability.
- Rituximab may be preferable for patients who value avoiding chronic medication, or who have a contraindication to, are unable to access, or cannot tolerate thrombopoietic agents.
- If rituximab is used, the most studied regimen is 375 mg/m2 × 4 weekly doses. However, other doses and schedules of rituximab may be used according to local practice. Biosimilars to rituximab are suitable substitutes.

Recommendation 2b. For adults with primary ITP needing additional treatment after initial therapy with corticosteroids (with or without IVIG), who are either ineligible for or choose not to receive a thrombopoietic agent or rituximab, the ASH guideline panel *suggests* a BTK inhibitor, MMF, or a SYK inhibitor (conditional recommendation based on very low certainty in the evidence ⨁◯◯◯).

Remarks:

- The panel concluded that current evidence in the population of interest is too indirect to definitively rank BTK inhibitors, MMF, or SYK inhibitors. Side effect profiles, cost, availability, and shared decision-making may be considered in selecting an agent.
- The panel concluded that azathioprine has a less favorable benefit to harm ratio; therefore, it is not equivalent to the other suggested treatment options. Nevertheless, azathioprine may be useful in some settings due to its low cost, global availability, and safety profile in pregnancy.

Background

Although the guideline panel suggests a combination of either rituximab or a thrombopoietic agent plus corticosteroids (with or without IVIG) as initial therapy (see Recommendation 1), it acknowledges that some patients will still be treated with corticosteroids (with or without IVIG) alone as initial therapy. The panel therefore prioritized the following question: In adults with primary ITP needing additional treatment after initial therapy with corticosteroids (with or without IVIG) alone, should an anti-CD20 agent, azathioprine, a BTK inhibitor, MMF, a SYK inhibitor, or a thrombopoietic agent be used, or should corticosteroids (with or without IVIG) be continued?

Summary of the evidence

This multi-comparison question compared six different interventions (MMF, thrombopoietic agent, anti-CD20, SYK inhibitor, BTK inhibitor, or azathioprine) to a uniform standard of care (SOC), which was defined as corticosteroids with or without IVIG. **Supplement 8a** presents the characteristics of all included studies for PICO 2 for MMF, thrombopoietic agent, anti-CD20, SYK inhibitor, BTK inhibitor, and azathioprine. **Supplement 8b** presents study level data for thrombopoietic and anti-CD20 agents for SRoT, durable response and HRQoL outcomes.

MMF

We found 1 prospective one-armed study [38] that addressed this question. The study included adults with primary ITP needing additional therapy after treatment with corticosteroids (with or without IVIG) and was considered indirect evidence since 15% of the study participants were splenectomized. The prospective one-armed study reported SRoT at 6 months, durable response at 6 months, and overall GI symptoms[38].

The evidence profile and EtD framework for this intervention is available online at https://guidelines.ash.gradepro.org/profile/wTP6UBZ2lXQ.

Thrombopoietic agents

We found 14 studies, 3 RCTs[39-41], 1 retrospective cohort study[42], 3 RCTs treated as prospective one-armed studies due to crossover at 6-10 weeks[43-45], 6 prospective one-armed studies[31,46-50], and 1 retrospective one-armed study[51], that addressed this question. Three studies also had an ancillary publication that contained prioritized outcomes (Primary/Ancillary: Kuter 2008[39]/George 2009[52]; Kuter 2010[40]/Kuter 2012[53]; Mei 2021[44]/Mei 2022[54]). Four studies were classified as direct evidence[42,47,48,50] since no patients had been splenectomized and 10 studies were classified as indirect evidence[31,39-41,43-46,49,51] since up to 20% of patients were splenectomized. Since our population of interest was patients who had previously received only initial therapy (with corticosteroids +/- IVIG) and not patients who had received multiple prior lines of therapy, we excluded studies with greater than or equal to 20% of patients who had been splenectomized (with no ability to separate splenectomized and non-splenectomized patients). The data from studies without a comparator arm were reported in the evidence tables, assessed by the panel, and informed the recommendations, but are only reported in the text if there was no evidence from controlled studies.

Four studies reported SRoT at 6 months[46-48,51], 3 studies reported SRoT at 12 months[48-50], *5* studies assessed durable response at 6 months[31,39,42-44], 3 studies assessed durable response at 12 months[40,49,54], 6 studies assessed major bleeding[31,41,43,48,50,54], 8 studies assessed minor bleeding[31,40,41,43,44,46,47,50], 4 studies assessed infection[40,41,44,45], 3 studies assessed overall GI symptoms[40,43,44], 6 studies assessed thrombosis[31,40,41,43,44,47], 4 studies assessed nausea[31,44,47,50], 2 studies assessed vomiting[31,50], 2 studies reported HRQoL[52,53], 1 study reported fatigue[31], 1 study reported diarrhea[47], 6 studies reported need for rescue therapy[39,41,43-46], and 4 studies reported time to response[42,44,46,49].

The evidence profile and EtD framework for this intervention is available online at https://guidelines.ash.gradepro.org/profile/hSqNgVFmvAg.

Anti-CD20

We found 5 studies, 3 RCTs[55-57], 1 retrospective cohort study[42], and 1 prospective one-armed cohort study[58], that addressed this question. All studies included adults with a diagnosis of primary ITP for 6 months or longer, needing additional therapy after treatment with corticosteroids (with or without IVIG or platelet transfusions). Three studies were classified as direct evidence[42,56,58] since no patients had been splenectomized and 2 studies were classified as indirect evidence[55,57] since up to 20% of patients were splenectomized . Studies with greater than or equal to 20% of patients who had been splenectomized (with no ability to separate splenectomized and non-splenectomized patients) were excluded.

One study reported SRoT at 6 months[55], 2 studies reported SRoT at 12 months[55,58], 3 studies assessed durable response at 6 months[42,56,57], 1 study assessed durable response at 12 months[56], 2 studies assessed major bleeding[55,57], 1 study assessed minor bleeding[56], 2 studies assessed thrombosis[56,58], 4 studies assessed infection[55-58], 2 studies assessed overall GI symptoms[57,58], 1 study reported need for rescue therapy[57], 1 study reported fatigue[57] and 2 studies reported time to response[42,58].

The evidence profile and EtD framework for this intervention is available online at https://guidelines.ash.gradepro.org/profile/jHfWCOEkAqI.

SYK Inhibitor

We found 4 studies, 1 one-armed post-hoc analysis of 2 RCTS[59] and 3 retrospective one-armed studies [60-62], that addressed this question. All studies included adults with a diagnosis of primary ITP for 6 months or longer, needing additional therapy after treatment with corticosteroids (with or without IVIG or platelet transfusions). One study was classified as direct evidence[59] since no patients had been splenectomized and 3 studies were classified as indirect evidence[60-62] since up to 20% of patients were splenectomized. Studies including 20% or more patients who had been splenectomized (with no ability to separate splenectomized and non-splenectomized patients) were excluded.

One study reported SRoT at 6 months[60], 4 studies assessed durable response at 6 months[59-62], 2 studies assessed durable response at 12 months[59,61], 1 study assessed minor bleeding[60], 1 study assessed thrombosis[60], 3 studies assessed diarrhea[60-62], 2 studies assessed time to response[60,61], 2 studies reported need for rescue therapy[59,61], 1 study reported depression[60], and 1 study reported nausea[60].

The evidence profile and EtD framework for this intervention is available online at https://guidelines.ash.gradepro.org/profile/XraioI1rAYY.

BTK Inhibitors

We found 4 studies: 1 RCT with a SOC comparison group[63], 1 RCT comparing doses with no SOC comparison group (dose groups combined and reported together) [64], 1 dose finding phase 1/2 clinical trial[65], and 1 prospective one-armed cohort study[66] that addressed this question. All studies included adults with a diagnosis of primary ITP for 6 months or longer, needing additional therapy after treatment with corticosteroids (with or without IVIG or platelet transfusions). No studies were classified as direct evidence and 4 studies were classified as indirect evidence [63-66]since up to 20% of patients were splenectomized. Studies including 20% or more patients who had been splenectomized (with no ability to separate splenectomized and non-splenectomized patients) were excluded.

One study reported SRoT at 6 months[66], 2 studies assessed durable response at 6 months[63,64], 1 study assessed major bleeding[66], 3 studies assessed minor bleeding[64-66], 2 studies assessed thrombosis[64,65], 3 studies assessed infection[64-66], 1 study assessed nausea[66], 2 studies assessed diarrhea[65,66], 1 study assessed vomiting[66], 2 studies assessed HRQoL[64,65], 3 studies reported need for rescue therapy[64-66], and 2 studies reported time to response[64,66].

The evidence profile and EtD framework for this intervention is available online at https://guidelines.ash.gradepro.org/profile/ov13DGGMco8.

Azathioprine

We found 5 studies: 1 RCT with a SOC comparator[26], 1 retrospective cohort study[67], 1 RCT with non SOC comparators[68] (only azathioprine arm reported), and 2 retrospective one-armed studies[69,70] that addressed this question. All studies included adults with a diagnosis of primary ITP for 6 months or longer, needing additional therapy after treatment with corticosteroids (with or without IVIG or platelet transfusions). Three studies were classified as direct evidence[26,67,69] since no patients had been splenectomized and 2 studies were classified as indirect evidence[68,70] since up to 20% of patients were splenectomized. Studies including 20% or more patients who had been splenectomized (with no ability to separate splenectomized and non-splenectomized patients) were excluded.

Two studies reported SRoT at 6 months[26,69], 2 studies assessed durable response at 6 months[67,68], 1 study assessed major bleeding[70], 2 studies assessed minor bleeding[26,70], 1 study assessed thrombosis[26], 4 studies assessed infection[26,67,69,70], 1 study assessed overall GI symptoms[70], 2 studies assessed nausea[67,68], 1 study assessed diarrhea[68], 1 study assessed anxiety[26], 1 study assessed depression[26], 1 study reported fatigue[26], 1 study reported need for rescue therapy[26], and 1 study reported time to response[70].

The evidence profile and EtD framework for this intervention is available online at https://guidelines.ash.gradepro.org/profile/gV_jlnhKT44.

Benefits

MMF

Only 1 indirect, single arm prospective study was identified. The reported SRoT at 6 months was 9 of 20 (45%) and durable response at 6 months was 13 of 20 (65%) for MMF. The panel judged the desirable effects of MMF to be small because there was only one eligible study, which did not include a control arm. The desirable effects from PICO 1 (increased durable response) were used as indirect evidence to assist in decisions related to desirable effects of this agent. Overall, the certainty of these estimated effects was very low due to serious inconsistency, and serious imprecision related to the inclusion of only one prospective single arm study. There was also serious indirectness since 15% of patients were splenectomized.

Thrombopoietic agents

The studies, both direct and indirect combined, showed a SRoT at 6 months of 30% (59 of 195) and an SRoT at 12 months of 27% (43 of 158). Thrombopoietic agents overall increased the odds of durable response at 6 months (direct – 10 of 11 (91%) thrombopoietic agent versus 34 of 41 (83%) SOC, OR: 2.06, 95% CI: 0.23 to 18.78);(indirect – 27 of 41 (66%) thrombopoietic agent versus 3 of 21 (14%) SOC, OR: 11.57, 95% CI: 2.90 to 46.10) and durable response at 12 months (indirect – 139 of 157 (89%) thrombopoietic agent versus 54 of 77 (70%) SOC, OR: 3.29, 95% CI: 1.65 to 6.57). Thrombopoietic agents decreased the odds of need for rescue therapy (indirect: 10 of 89 (11%) thrombopoietic agent versus 17 of 45 (38%) SOC, OR: 0.18, 95% CI: 0.07 to 0.47). The studies found no meaningful difference between those receiving a thrombopoietic agent and control groups for minor bleeding (indirect 97 of 202 (48%) thrombopoietic agent versus 48 of 99 (49%) SOC, OR: 0.94, 95% CI: 0.47 to 1.89), overall GI symptoms (indirect - 18 of 202 (9%) thrombopoietic agent versus 8 of 101 (8%) SOC, OR: 0.86, 95% CI: 0.11 to 6.57), or nausea (indirect - 3 of 48 (6%) thrombopoietic agent versus 1 of 26 (4%) SOC, OR: 1.67, 95% CI: 0.16 to 16.88). Time to response may be longer for thrombopoietic agents as compared to the SOC (direct – median 1.5 days longer for thrombopoietic agent) and HRQoL may be greater for thrombopoietic agent (indirect: mean difference 3.67 higher (95% CI: -5.06 to 12.39)), but the estimates were imprecise and the CIs include no effect. The panel judged the desirable effects of thrombopoietic agents to be large, mainly driven by durable response at 6 and 12 months, and SRoT at 6 and 12 months. Additionally, there was a reduction in need for rescue therapy, and improvements in various HRQoL domains. The panel noted that duration of treatment with a thrombopoietic agent was variable as was dosing among included studies. Overall, the certainty of these estimates was deemed moderate based on three critically important outcomes (durable response at 6 months, durable response at 12 months, and need for rescue therapy).

Anti-CD20

Anti-CD20 increased the odds of SRoT at 6 months: (indirect: 24 of 31 (77%) anti-CD20 versus 12 of 31 (39%) SOC, OR: 5.43, 95% CI: 1.79 to 16.47) and SRoT at 12 months (indirect:21 of 31 (68%) anti-CD20 versus 6 of 31 (19%) SOC, OR: 8.75, 95% CI: 2.72 to 28.10). Anti-CD20 may increase the odds of durable response at 6 months (direct: 79 of 90 (88%) anti-CD20 versus 74 of 95 (78%) SOC; OR: 1.12, 95% CI: 0.95 to 1.28; indirect: 20 of 32 (63%) anti-CD20 versus 19 of 26 (73%) SOC; OR: 0.86, 95% CI: 0.60 to 1.25), and durable response at 12 months (direct: 35 of 55 (64%) anti-CD20 versus 28 of 54 (52%) SOC; OR: 1.63, 95% CI: 0.76 to 3.60) and may decrease the odds, need for rescue therapy (14 of 32 (44%) anti-CD20 versus 17 of 26 (65%) SOC, OR: 0.41, 95% CI: 0.14 to 1.20), overall GI symptoms (direct: 0 of 32 (0%) anti-CD20 versus 4 of 26 (15%) SOC; OR: 0.08, 95% CI: 0.00 to 1.50), and minor bleeding (21 of 55 (38%) anti-CD20 versus 27 of 54 (59%) SOC, OR: 0.62, 95% CI: 0.29 to 1.32) but the estimates were imprecise and the CIs include no effect. The panel judged the desirable effects of anti-CD20 agents to be small, driven primarily by SRoT with rituximab at 6 and 12 months, though the panel noted that SRoT was likely a surrogate for durable response given that small amounts of treatment (e.g. prednisone <7.5mg daily) were permitted in the largest randomized trial[56]. The panel noted that the effect of anti-CD20 on B-cell depletion can last for many months; thus 'exposure' to the pharmacodynamic effects of the drug persists well after treatment is completed. Overall, the certainty of these estimates is deemed low due to serious inconsistency, serious imprecision, and serious indirectness. The decision to assign low certainty of evidence was also influenced by differences in anti-CD20 dosing.

SYK Inhibitor

Only single arm studies were identified. The reported SRoT at 6 months was 4 of 122 (3%), durable response at 6 months was 125 of 238 (53%), and durable response at 12 months was 24 of 63 (38%). The panel judged the desirable effects of SYK inhibitors to be small, driven primarily by durable response at 6 and 12 months. Some studies included refractory patients. The panel noted that because there was efficacy in a more refractory (indirect) patient population, response in the population of interest would be expected to be at least as high. Overall, the certainty of these estimated effects was very low due to very serious to serious risk of bias, serious inconsistency, serious indirectness, and single arm study design.

BTK Inhibitor

BTK inhibitors increased the odds of durable response at 6 months (26 of 96 (27%) BTK versus 0 of 50 (0%) SOC, OR: 37.96, 95% CI: 2.26 to 637.61) but the CI was very wide. The reported SRoT at 6 months was 7 of 20 (35%). Increases in HRQoL were also reported (average increase of 21.2 points in physical well-being and 10.3 points in emotional well-being using the 36-Item Short Form Health Survey (SF-36) scores measured at week 24). The panel judged the desirable effects of BTK inhibitors to be small, driven primarily by increased durable response at 6 months. Some studies included refractory patients. The panel noted that because there was efficacy in a more refractory (indirect) patient population, response in the population of interest would be expected to be at least as high. Overall, the certainty of these estimated effects was very low due to very serious to serious risk of bias, serious inconsistency, serious indirectness, and single arm study design.

Azathioprine

Azathioprine (AZA) decreased the odds of anxiety (direct: 86 of 110 (78%) AZA versus 94 of 105 (90%) SOC OR: 0.42, 95% CI: 0.19 to 0.91). Azathioprine may increase the odds of durable response at 6 months (direct: 10 of 15 (67%) AZA versus 9 of 18 (50%) SOC; OR: 2.00, 95% CI: 0.49 to 8.24), decreased the odds of depression (direct: 93 of 110 (85%) AZA versus 99 of 105 (95%) SOC OR: 0.33, 95% CI: 0.193 to 0.88), and may decrease fatigue (direct: 74 of 110 (67%) AZA versus 81 of 105 (77%) SOC OR: 0.61, 95% CI: 0.33 to 1.12), but the estimates were imprecise and CIs include no effect. Studies found no meaningful difference between those receiving azathioprine and SOC for the need for rescue therapy (direct: 19 of 110 (17%) AZA versus 15 of 105 (14%) SOC OR: 1.25, 95% CI: 0.16 to 2.62). The reported percentage of patients experiencing major bleeding who received azathioprine was low (1 of 63 (2%). The panel judged the desirable effects of azathioprine to be trivial, driven primarily by small improvement in durable response at 6 months and improvements in various domains of HRQoL. The panel noted that many patients in the included studies were refractory (indirect) and that response rates would be expected to be at least as high in the population of interest. Overall, the certainty of these estimated effects was very low due to serious risk of bias, serious inconsistency, and serious indirectness.

Harms and burden

MMF

For MMF, overall GI symptoms occurred in 7 of 20 (35%). There was very low certainty in the estimate of the risk of adverse effects due to serious inconsistency, and serious imprecision related to the inclusion of only one prospective single arm study. There was also serious indirectness since 15% of patients were splenectomized. However, given the available evidence, the guideline panel considered the risk of adverse effects most likely to be small, primarily driven by GI symptoms. Some panel members noted the undesirable effects could be considered moderate. The undesirable effects were extrapolated from an indirect population; however, undesirable effects were not expected to be substantially different in the population of interest.

Thrombopoietic agents

Thrombopoietic agents may increase the odds of infection (23 of 202 (11%) thrombopoietic agent versus 5 of 101 (5%) SOC, OR: 2.43, 95% CI: 0.49 to 12.05) and thrombosis (indirect: 6 of 202 (3%) thrombopoietic agent versus 2 of 101 (2%) SOC; OR: 1.48, 95% CI: 0.29 to 7.51) but the estimates were imprecise and the CIs include no effect. There was low certainty in the estimate of the risk of adverse effects due to serious indirectness, and serious imprecision. However, given the available evidence, the guideline panel considered the risk of adverse effects most likely to be small, primarily driven by thrombosis. Patients may be adversely impacted by dietary restrictions associated with certain thrombopoietic agents (i.e., eltrombopag) or the need for weekly injections in a healthcare setting (i.e., romiplostim). The panel also noted other side effects that were not prioritized including musculoskeletal pain and liver injury (with eltrombopag).

Anti-CD20

Anti-CD20 may increase the odds of thrombosis (direct: 2 of 55 (4%) anti-CD20 versus 0 of 54 (0%) SOC, OR: 5.09, 95% CI: 0.24 to 108.60), infection (direct: 22 of 55 (40%) anti-CD20 versus 13 of 54 (24%) SOC, OR: 2.10, 95% CI: 0.92 to 4.80; indirect: 7 of 63 (11%) anti-CD20 versus 7 of 57 (12%), OR: 1.26, 95% CI: 0.38 to 4.23), fatigue (7 of 32 (22%) anti-CD20 versus 2 of 26 (8%) SOC, OR: 3.36, 95% CI: 0.63 to 17.82), and major bleeding (9 of 63 (14%) anti-CD20 versus 6 of 57 (11%) SOC, OR: 1.25, 95% CI: 0.34 to 24.52) but the estimates were imprecise and the CIs include no effect. Time to response was a median of 1.5 days longer in anti-CD20 arm compared to SOC. There was low certainty in the estimate of the risk of adverse effects due to serious inconsistency, serious imprecision, and serious indirectness. However, given the available evidence, the guideline panel considered the risk of adverse effects to be small, primarily driven by increased infection. The panel noted that serious infections (e.g., progressive multifocal leukoencephalopathy) are very rare in ITP, but have been reported in other clinical settings, typically when anti-CD20 agents were given in combination with other immunosuppressive agents. The panel noted that treatment with anti-CD20 attenuates the response to vaccinations for at least 6 months. The panel noted that there are other important adverse effects that were not prioritized such as hepatitis B reactivation, neutropenia, serum sickness, and infusion reactions.

SYK Inhibitor

Only single arm studies were identified. The reported need for rescue therapy was 28 of 63 (44%), minor bleeding 20 of 138 (15%), and diarrhea 38 of 209 (18%). The reported risk of thrombosis (2 of 138 (1%)), depression (2 of 138 (1%)) and nausea (1 of 138 (1%)) was low. The panel judged the undesirable effects of SYK inhibitors to be small, primarily driven by diarrhea. Additionally, there was a low percentage of patients who reported thrombosis. Panel members discussed that the undesirable effects could be considered moderate; however, they settled on small because most toxicities were tolerable and mild and responded to symptomatic treatment. Other unprioritized side effects noted by the panel include hypertension, neutropenia, and liver injury.

BTK Inhibitor

Only single arm studies were identified. The adverse events of thrombosis (1 of 49 (2%)) and vomiting (1 of 20 (5%)) were reported 5% or less of the time. The reported rate of major bleeding was 2 of 20 (10%), minor bleeding was 9 of 69 (13%), infection was 14 of 69 (20%), nausea was 2 of 20 (10%), diarrhea was 6 of 36 (17%), need for rescue therapy was 23 of 69 (33%), and time to response was a median of 14 days. The panel judged the undesirable effects of BTK inhibitors to be small. This was primarily driven by need for rescue therapy, infection, diarrhea, and minor bleeding.

Azathioprine

Azathioprine increased the odds of minor bleeding (direct: 86 of 110 (78%) AZA versus 62 of 105 (59%) SOC; OR: 2.49 95% CI: 1.37 to 4.51). Azathioprine may decrease the odds of SRoT at 6 months (direct: 13 of 72 (18%) AZA versus 23 of 79 (29%) SOC; OR:0.54, 95% CI: 0.25 to 1.16), but the estimate is imprecise and the confidence interval includes no effect. Studies found no meaningful difference between those receiving azathioprine and SOC for the outcomes of thrombosis: (direct 0 of 110 (0%) versus 0 of 105 (0%) – not pooled), infection (direct 15 of 125 (12%) AZA versus 18 of 123 (15%) SOC; OR: 0.78 95% CI:

0.37 to 1.64), and nausea (direct: 1 of 15 (7%) AZA versus 1 of 18 (6%) SOC, OR: 1.21 95% CI: 0.07 to 21.22). There is very low certainty in the estimate of the risk of adverse effects due to serious risk of bias, serious inconsistency, and serious indirectness. However, given the available evidence, the guideline panel considered the risk of adverse effects most likely to be moderate. This was primarily driven by the delayed time to response (up to 3 months) and reduced SRoT at 6 months relative to the comparator arm. Additionally, panel members discussed that some patients do not tolerate this medication well, and that treatment can be complicated by drug-drug interactions. Important potential side effects that were not prioritized include cytopenias, transaminitis, myalgias, and rash.

Other EtD criteria and considerations

The guideline panel noted that the cost of MMF and azathioprine is negligible, the cost of rituximab is moderate, and the cost of BTK inhibitors, SYK inhibitors, and thrombopoietic agents is large. The panel noted that the cost of thrombopoietic agents is highly variable depending on which agent is used as well as dose, duration, and jurisdiction. The panel emphasized that approximately 30% of patients treated with a thrombopoietic agent are ultimately able to achieve a SRoT, which mitigates the cost of this class of medications[71].

The panel determined that BTK inhibitors, SYK inhibitors, thrombopoietic agents, and rituximab would reduce or probably reduce equity due to cost, uneven insurance coverage, and lack of availability in certain jurisdictions for some agents. They noted that MMF would probably reduce equity because it is contraindicated in pregnancy. Conversely, they emphasized that azathioprine is inexpensive, available globally, and can be used safely in pregnancy.

The panel judged that BTK inhibitors, SYK inhibitors, MMF, rituximab, and thrombopoietic agents are probably acceptable and probably feasible, with cost and insurance coverage being the most significant barriers, particularly for BTK inhibitors, SYK inhibitors, and thrombopoietic agents. The panel judged azathioprine to be probably acceptable and feasible given its low cost and broad availability.

Conclusions for this recommendation

Results of the panel discussion of the pairwise comparisons informed the multi-comparison EtD that was used to form final recommendations The shared multicomparison EtD for PICO 2 is available in the "Evidence to Decision framework" tab of the relevant agent-specific evidence profiles linked in each agent's subsection in "Summary of the evidence."

The panel issued a strong recommendation in favor of thrombopoietic agents, primarily driven by large benefits in the form of greater SRoT and durable response at 6 and 12 months, decreased need for rescue therapy, and improvement in various dimensions of HRQoL. These desirable effects were considered to more than offset the small undesirable effects (driven mainly by thrombotic risk), large costs, and probably reduced equity associated with this class of agents.

For patients who place a high value on avoiding chronic medication, or who have a contraindication to, are unable to access, or cannot tolerate thrombopoietic agents, the panel issued a conditional recommendation in favor of rituximab. The panel judged the benefits of rituximab to be small (driven mainly by greater SROT at 6 and 12 months) and determined that the benefits outweigh the small harms (driven mainly by infection), moderate costs, and probable reduction in equity associated with this agent.

For patients who are unable or unwilling to be treated with a thrombopoietic agent or rituximab, the panel issued a conditional recommendation in favor of a BTK inhibitor, SYK inhibitor, or MMF. Although there is more abundant evidence with these agents in patients with chronic and/or more heavily pretreated ITP, the panel determined that evidence in the population of interest is insufficient to rank order them and noted that consideration of side effect profiles, cost, and availability may be useful in selecting an agent through a shared decision-making approach.

These considerations support an individualized approach to selection of second-line therapy for adults with primary ITP corresponding to Recommendation 2 and incorporating patient values, preferences, and treatment availability, as summarized in Figure 3.

Finally, the panel concluded that azathioprine has a less favorable benefit to harm ratio and should therefore be deprioritized compared with the other options. This conclusion was based on the panel's judgement that the moderate undesirable effects of azathioprine (driven mainly by delayed time to response, reduced SRoT at 6 months, and well-known toxicity profile) outweigh its trivial benefits (driven mainly by small increased durable response at 6 months). Nevertheless, the panel noted that azathioprine may be useful in some settings owing to its low cost, global availability, and safety profile in pregnancy.

Good Practice Statements

Splenectomy

Background: Splenectomy is an appropriate consideration in adults with primary ITP in the following circumstances: 1) patients who are intolerant to or have failed multiple lines of medical therapy at optimal doses and durations, 2) patients who wish to avoid long-term medical treatment, and 3) patients with severe thrombocytopenia who need emergent treatment and have an inadequate response to medical therapies.

Indium-labeled autologous platelet scintigraphy (ILAPS) predicts the likelihood of response to splenectomy. In a systematic review and meta-analysis, the response rate was 87.1%, 47.1%, and 25.5% in patients with a splenic, mixed, and hepatic sequestration pattern, respectively[72]. ILAPS is currently only offered at a small number of centers around the world.

Normalization of the platelet count following splenectomy does not prevent transplacental passage of maternal anti-platelet antibodies during pregnancy, which can cause neonatal thrombocytopenia.

Good practice statement: If feasible, splenectomy should be delayed for at least one year after diagnosis because of the potential for remission.

Good practice statement: The treating clinician should ensure that patients receive appropriate immunizations at least two weeks prior to splenectomy, when feasible, and counseling regarding antibiotic prophylaxis and lifelong risk of sepsis and thrombosis following splenectomy.
The treating clinician should educate the patient on prompt recognition and management of fever with antibiotics. After splenectomy, patients should be followed at least annually for platelet count monitoring, counseling, and to ensure adherence to recommended immunization schedules.

Thrombopoietic agent switching

Background: Patients differ in their response to and tolerance of individual thrombopoietic agents. Patients who have failed to respond to one agent may respond to a different one[73].

Good practice statement: Switching to a different thrombopoietic agent may be considered based on insufficient response, adverse effects, comorbidities, poor adherence, or patient preference.

Good practice statement: Shared decision-making with the patient, documentation of the rationale for switching, avoidance of treatment gaps, and close monitoring around the time of switching are good practices to ensure optimal outcomes and continuity of care.

What are others saying and what is new in these guidelines?

The 2026 ASH ITP guidelines include both methodologic innovations and novel recommendations. In recognition of the growing number of treatment options available for ITP, these guidelines are the first to use the GRADE evidence to decision framework for multiple comparisons[18], whereas past ITP guidelines including the 2019 ASH ITP guideline were limited to pairwise comparisons[15].

With respect to initial therapy, previously published guidelines including the 2019 ASH ITP guidelines, the 2019 International Consensus Report, and national guidelines advocate for corticosteroids alone in the first-line setting[15,74-76]. In contrast, our guidelines suggest rituximab or a thrombopoietic agent plus corticosteroids as initial therapy (see Recommendation 1), a recommendation driven mainly by greater SRoT at 6 and 12 months, reduced need for rescue therapy, and improvement in various dimensions of HRQoL with combination therapy.

Among patients who require treatment beyond an initial course of corticosteroids, recommendation 2 in these guidelines ranks second-line therapy options as follows: (1) thrombopoietic agents, (2) rituximab, (3) BTK inhibitor or MMF or SYK inhibitor, (4) azathioprine. As a recently approved treatment for ITP in 2025, BTK inhibitors are not addressed in previously published guidelines. The other treatment options are addressed but not explicitly rank-ordered in other guidelines or consensus documents[15,74-76].

Limitations of these guidelines

A key limitation of these guidelines is inherent in the low or very low certainty of the evidence we identified for many of the questions and interventions. Indeed, we were able to issue only one strong recommendation, highlighting the need for further randomized controlled trials comparing different active treatment strategies. For some questions and interventions, the panel relied on indirect evidence, especially when direct evidence was lacking. We specify in the text where indirect evidence was applied.

Another key limitation of these guidelines is their restricted scope. We undertook a focused revision of the 2019 ASH ITP guidelines, specifically centered on management of adults with primary ITP requiring initial or second line therapy. Other important aspects of ITP including diagnosis, indications for treatment, management of children, management of ITP in pregnancy, management of secondary ITP, management of chronic and/or refractory ITP, and management of critical bleeding are addressed elsewhere.[15,74,77]

In accordance with the GRADE framework for multiple comparisons, we were limited to no more than 7 interventions per question. The guideline panel prioritized which interventions to include, but was not able to consider some of the many ITP treatments used around the world including cyclosporine, danazol, dapsone, sirolimus, tacrolimus, and others.

Finally, there may be confusion about how Recommendations 1 and 2 align. Recommendation 1 suggests initial treatment with either rituximab or a thrombopoietic agent plus corticosteroids (with or without IVIG) rather than corticosteroids (with or without IVIG) alone. Notwithstanding this recommendation, the guideline panel acknowledged that many clinicians will continue to prescribe corticosteroids (with or without IVIG) alone as initial therapy. Recommendation 2 provides guidance on how to manage such patients if they require treatment beyond corticosteroids.

Revision or adaptation of the guidelines

Plans for updating these guidelines

After publication of these guidelines, ASH will maintain them through surveillance for new evidence, ongoing review by experts, and regular revisions.

Updating or adapting recommendations locally

Adaptation of these guidelines will be necessary in many circumstances. These adaptations should be based on the associated EtD frameworks[78].

Priorities for research

On the basis of gaps in evidence identified during the guideline development process, the panel makes the following recommendations for further research:

- Determination of the optimal thrombopoietic agent, dose, and duration of treatment for initial therapy
- Determination of the optimal rituximab dose and schedule for initial therapy
- Randomized controlled trials comparing rituximab plus corticosteroids (with or without IVIG) vs. a thrombopoietic agent plus corticosteroids (with or without IVIG) as initial therapy
- Long-term follow-up to determine SRoT beyond 12 months after initial combination therapy. Do these regimens "cure" patients or do they merely postpone relapse?
- Studies of other potential combination regimens for initial therapy such as SYK inhibitors, BTK inhibitors, BAFF receptor inhibitors, or anti-CD38 agents plus corticosteroids (with or without IVIG)
- Studies of the interval between diagnosis and commencement of initial therapy and its potential impact on outcomes
- Studies of corticosteroid regimens other than high-dose dexamethasone as the corticosteroid component of initial combination therapy
- More and higher quality evidence on second-line treatment with azathioprine, BTK inhibitors, MMF, and SYK inhibitors
- Clinical trials of other agents in the second-line setting including anti-CD38 agents, BAFF receptor inhibitors, neonatal Fc receptor antagonists, and complement inhibitors
- Studies to define the optimal timing of initiation of second-line therapy
- Identification of clinical predictors and/or biomarkers to guide prioritization and sequencing of specific therapies in individual patients
- Evaluation of the efficacy and safety of ITP therapies in pregnancy
- Evaluation of azathioprine, BTK inhibitors, MMF, and SYK inhibitors in the refractory setting, both as monotherapy and as a component of combination therapy

Acknowledgments

The authors gratefully acknowledge Rachel Cohen and Rob Kunkle of the American Society of Hematology for supporting the guideline panel and its efforts.

The authors gratefully acknowledge Jessica Reese, PhD for her contributions to supporting the systematic literature review and meta-analysis.

The authors and ASH staff gratefully acknowledge the Platelet Disorder Support Association (PDSA) for its contributions in identifying patient and caregiver representatives who informed the development of these guidelines.

Authorship contributions

AC, DRT, and SKV wrote the first draft of this manuscript and revised the manuscript based on authors' suggestions. Guideline panel members (DMA, SA, JBB, TJG-L, RFG, CM, KRM, MM, CN, SRP, SS, BS, JV) critically reviewed the manuscript and provided suggestions for improvement. Members of the knowledge synthesis team (DRT, HS, SKV) contributed evidence summaries to the guidelines. All authors approved of the content. DBC and SKV were the co-chairs of the panel and led panel meetings.

Disclosures of conflicts of interest

All authors were members of the guideline panel or members of the systematic review team or both. As such, they completed a disclosure of interest form, which was reviewed by ASH and is available as supplements 2 and 3.

Dedication

The 2026 ASH ITP guidelines are dedicated to Dr. Douglas Cines, who served as Chair of the guideline panel and provided invaluable contributions and inspiration to this work until his untimely death in December 2025. Dr. Cines was a brilliant clinician and scientist and a cherished colleague, mentor, and friend.

Correspondence

Adam Cuker, Hospital of the University of Pennsylvania, 3400 Spruce Street, Philadelphia, PA 19104; Adam.Cuker@pennmedicine.upenn.edu.

Figures

Figure 1. Interpretation of Recommendations

Figure 2. Interpretation of Certainty of Evidence Ratings

Figure 3. Treatment Pathways for Adults with Primary ITP

Interpretation of the treatment pathways for adults with primary ITP. Arrows indicate the sequence of clinical decisions, beginning with assessment of patient values and eligibility and proceeding to suggested treatments. Red header bars define the clinical scenario (initial vs additional therapy), orange boxes show specific treatment options or actions, and purple boxes highlight good practice statements. Green checkmarks denote suggested or recommended options, while the symbols at the bottom of each treatment box indicate strength of recommendation and certainty of evidence as defined in the key.

**Although the guideline panel suggests a combination of either a thrombopoietic agent or rituximab plus corticosteroids (with or without IVIG) as initial therapy (Recommendation 1), it acknowledges that some patients will still be treated with corticosteroids (with or without IVIG) alone as initial therapy. Recommendation 2 addresses management of such patients if treatment beyond initial therapy with corticosteroids is required.*

References

1. Institute of Medicine Committee on Standards for Developing Trustworthy Clinical Practice G. In: Graham R, Mancher M, Miller Wolman D, Greenfield S, Steinberg E, eds. *Clinical Practice Guidelines We Can Trust*. National Academies Press (US)

Copyright 2011 by the National Academy of Sciences. All rights reserved.; 2011.

2. Qaseem A, Forland F, Macbeth F, Ollenschläger G, Phillips S, van der Wees P. Guidelines International Network: toward international standards for clinical practice guidelines. *Ann Intern Med*. Apr 3 2012;156(7):525-31. doi:10.7326/0003-4819-156-7-201204030-00009

3. Schünemann HJ, Al-Ansary LA, Forland F, et al. Guidelines International Network: Principles for Disclosure of Interests and Management of Conflicts in Guidelines. *Ann Intern Med*. Oct 6 2015;163(7):548-53. doi:10.7326/m14-1885

4. Neumann I SH, ed. *The GRADE Book. The GRADE Working Group*. 2024. Available from: https://book.gradepro.org

5. Cines DB, Cuker A, Semple JW. Pathogenesis of immune thrombocytopenia. *Presse Med*. Apr 2014;43(4 Pt 2):e49-59. doi:10.1016/j.lpm.2014.01.010

6. Audia S, Mahévas M, Nivet M, Ouandji S, Ciudad M, Bonnotte B. Immune Thrombocytopenia: Recent Advances in Pathogenesis and Treatments. *Hemasphere*. Jun 2021;5(6):e574. doi:10.1097/hs9.0000000000000574

7. Frederiksen H, Schmidt K. The incidence of idiopathic thrombocytopenic purpura in adults increases with age. *Blood*. Aug 1 1999;94(3):909-13.

8. Terrell DR, Beebe LA, Vesely SK, Neas BR, Segal JB, George JN. The incidence of immune thrombocytopenic purpura in children and adults: A critical review of published reports. *Am J Hematol*. Mar 2010;85(3):174-80. doi:10.1002/ajh.21616

9. Abrahamson PE, Hall SA, Feudjo-Tepie M, Mitrani-Gold FS, Logie J. The incidence of idiopathic thrombocytopenic purpura among adults: a population-based study and literature review. *Eur J Haematol*. Aug 2009;83(2):83-9. doi:10.1111/j.1600-0609.2009.01247.x

10. Terrell DR, Beebe LA, Neas BR, Vesely SK, Segal JB, George JN. Prevalence of primary immune thrombocytopenia in Oklahoma. *Am J Hematol*. Sep 2012;87(9):848-52. doi:10.1002/ajh.23262

11. Weycker D, Hanau A, Hatfield M, et al. Primary immune thrombocytopenia in US clinical practice: incidence and healthcare burden in first 12 months following diagnosis. *J Med Econ*. Feb 2020;23(2):184-192. doi:10.1080/13696998.2019.1669329

12. Cuker A, Cines DB. Immune thrombocytopenia. *Hematology Am Soc Hematol Educ Program*. 2010;2010:377-84. doi:10.1182/asheducation-2010.1.377

13. Huang K. Novel therapeutics for immune thrombocytopenia: an evolving treatment landscape. *Expert Rev Hematol*. Jan 2026;19(1):17-31. doi:10.1080/17474086.2025.2570332

14. Dewidar O, Lotfi T, Langendam MW, et al. Good or best practice statements: proposal for the operationalisation and implementation of GRADE guidance. *BMJ Evid Based Med*. Jun 2023;28(3):189-196. doi:10.1136/bmjebm-2022-111962

15. Neunert C, Terrell DR, Arnold DM, et al. American Society of Hematology 2019 guidelines for immune thrombocytopenia. *Blood Adv*. Dec 10 2019;3(23):3829-3866. doi:10.1182/bloodadvances.2019000966

16. Neunert CE, Arnold DM, Grace RF, Kuhne T, McCrae KR, Terrell DR. The 2022 review of the 2019 American Society of Hematology guidelines on immune thrombocytopenia. *Blood Adv*. Jul 9 2024;8(13):3578-3582. doi:10.1182/bloodadvances.2023012541

17. Cines DB, Bussel JB, Liebman HA, Luning Prak ET. The ITP syndrome: pathogenic and clinical diversity. *Blood*. Jun 25 2009;113(26):6511-21. doi:10.1182/blood-2009-01-129155

18. Piggott T, Brozek J, Nowak A, et al. Using GRADE evidence to decision frameworks to choose from multiple interventions. *J Clin Epidemiol*. Feb 2021;130:117-124. doi:10.1016/j.jclinepi.2020.10.016

19. *GRADEpro Guideline Development Tool (G2DT) (Computer Program)* Version 2.xx. 2014. www.guidelinedevelopment.org

20. Alonso-Coello P, Oxman AD, Moberg J, et al. GRADE Evidence to Decision (EtD) frameworks: a systematic and transparent approach to making well informed healthcare choices. 2: Clinical practice guidelines. *Bmj*. Jun 30 2016;353:i2089. doi:10.1136/bmj.i2089

21. Alonso-Coello P, Schünemann HJ, Moberg J, et al. GRADE Evidence to Decision (EtD) frameworks: a systematic and transparent approach to making well informed healthcare choices. 1: Introduction. *Bmj*. Jun 28 2016;353:i2016. doi:10.1136/bmj.i2016

22. Schünemann HJ, Mustafa R, Brozek J, et al. GRADE Guidelines: 16. GRADE evidence to decision frameworks for tests in clinical practice and public health. *J Clin Epidemiol*. Aug 2016;76:89-98. doi:10.1016/j.jclinepi.2016.01.032

23. Sterne JA, Hernán MA, Reeves BC, et al. ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. *Bmj*. Oct 12 2016;355:i4919. doi:10.1136/bmj.i4919

24. Sterne JAC, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *Bmj*. Aug 28 2019;366:l4898. doi:10.1136/bmj.l4898

25. Bradbury CA, Pell J, Hill Q, et al. Mycophenolate Mofetil for First-Line Treatment of Immune Thrombocytopenia. *New England Journal of Medicine*. 2021;385(10):885-895. doi:https://dx.doi.org/10.1056/NEJMoa2100596

26. Hamed EM, Ibrahim ARN, Meabed MH, et al. The Outcomes and Adverse Drug Patterns of Immunomodulators and Thrombopoietin Receptor Agonists in Primary Immune

Thrombocytopenia Egyptian Patients with Hemorrhage Comorbidity. *Pharmaceuticals*. 2023;16(6):16. doi:10.3390/ph16060868

27. Yu Y, Wang M, Hou Y, et al. High-dose dexamethasone plus recombinant human thrombopoietin vs high-dose dexamethasone alone as frontline treatment for newly diagnosed adult primary immune thrombocytopenia: A prospective, multicenter, randomized trial. *American Journal of Hematology*. 2020;95(12):1542-1552. doi:https://dx.doi.org/10.1002/ajh.25989

28. Dehghani-Ghorbi M, Azimi R, Taherpour N, Sadeghi S, Homaee S, Saberian F. Comparative analysis of three treatment approaches for primary immune thrombocytopenic purpura; prednisolone with thrombopoietin, prednisolone alone, and pulse methylprednisolone; a randomized clinical trial. *IMMUNOPATHOLOGIA PERSA*. 2025;11(2)doi:10.34172/ipp.2025.43817

29. Gomez-Almaguer D, Herrera-Rojas MA, Jaime-Perez JC, et al. Eltrombopag and high-dose dexamethasone as frontline treatment of newly diagnosed immune thrombocytopenia in adults. *Blood*. 2014;123(25):3906-8. doi:https://dx.doi.org/10.1182/blood-2014-01-549360

30. Zhang L, Zhang M, Du X, Cheng Y, Cheng G. Safety and efficacy of eltrombopag plus pulsed dexamethasone as first-line therapy for immune thrombocytopenia. *British Journal of Haematology*. 2020;189(2):369-378. doi:https://dx.doi.org/10.1111/bjh.16327

31. Reiser M, Josten KM, Dietzfelbinger H, et al. Romiplostim for Primary Immune Thrombocytopenia in Routine Clinical Practice: Results from a Multicentre Observational Study in Germany. *Acta Haematologica*. 2022;145(4):394-403. doi:https://dx.doi.org/10.1159/000521689

32. Tripathi AK, Shukla A, Mishra S, Yadav YS, Yadav DK. Eltrombopag therapy in newly diagnosed steroid non-responsive ITP patients. *International Journal of Hematology*. 2014;99(4):413-7. doi:https://dx.doi.org/10.1007/s12185-014-1533-y

33. Gudbrandsdottir S, Birgens HS, Frederiksen H, et al. Rituximab and dexamethasone vs dexamethasone monotherapy in newly diagnosed patients with primary immune thrombocytopenia. *Blood*. 2013;121(11):1976-81. doi:10.1182/blood-2012-09-455691

34. Zaja F, Baccarani M, Mazza P, et al. Dexamethasone plus rituximab yields higher sustained response rates than dexamethasone monotherapy in adults with primary immune thrombocytopenia. *Blood*. 2010;115(14):2755-62. doi:https://dx.doi.org/10.1182/blood-2009-07-229815

35. Gomez-Almaguer D, Tarin-Arzaga L, Moreno-Jaime B, et al. High response rate to low-dose rituximab plus high-dose dexamethasone as frontline therapy in adult patients with primary immune thrombocytopenia. *European Journal of Haematology*. 2013;90(6):494-500. doi:https://dx.doi.org/10.1111/ejh.12102

36. Zhou H, Liu L, Shu X, Wang X, Song Y. Clinical Efficacy and Safety of High-Dose Dexamethasone Plus Low-Dose Rituximab as First-Line Therapy in Newly Diagnosed Primary Immune Thrombocytopenia. *Indian Journal of Hematology and Blood Transfusion*. 2019;35(3):507-512. doi:https://dx.doi.org/10.1007/s12288-018-1061-7

37. Coronado-Alejandro EU, Martinez-Hernandez JP, Gomez-De Leon A, et al. Low-dose rituximab plus high-dose dexamethasone in newly diagnosed immune thrombocytopenia. *European Journal of Haematology*. 2023;110(6):778-779. doi:https://dx.doi.org/10.1111/ejh.13954

38. Zhang W-g, Ji L, Cao X-m, et al. Mycophenolate mofetil as a treatment for refractory idiopathic thrombocytopenic purpura. *Acta pharmacologica Sinica*. 2005;26(5):598-602. doi:https://dx.doi.org/10.1111/j.1745-7254.2005.00088.x

39. Kuter DJ, Bussel JB, Lyons RM, et al. Efficacy of romiplostim in patients with chronic immune thrombocytopenic purpura: a double-blind randomised controlled trial. *Lancet (London, England)*. 2008;371(9610):395-403. doi:https://dx.doi.org/10.1016/S0140-6736(08)60203-2

40. Kuter DJ, Rummel M, Boccia R, et al. Romiplostim or standard of care in patients with immune thrombocytopenia. *The New England journal of medicine*. 2010;363(20):1889-99. doi:https://dx.doi.org/10.1056/NEJMoa1002625

41. Mei H, Zhou H, Hou M, et al. Avatrombopag for adult chronic primary immune thrombocytopenia: a randomized phase 3 trial in China. *Research and Practice in Thrombosis and Haemostasis*. 2023;7(6):12. doi:10.1016/j.rpth.2023.102158

42. Jaime-Perez JC, Aguilar-Calderon P, Jimenez-Castillo RA, Ramos-Davila EM, Salazar-Cavazos L, Gomez-Almaguer D. Treatment outcomes and chronicity predictors for primary immune thrombocytopenia: 10-year data from an academic center. *Annals of Hematology*. 2020;99(11):2513-2520. doi:https://dx.doi.org/10.1007/s00277-020-04257-2

43. Zhou H, Wu D, Ma L, et al. Romiplostim in primary immune thrombocytopenia that is persistent or chronic: phase III multicenter, randomized, placebo-controlled clinical trial in China. *Research and Practice in Thrombosis and Haemostasis*. 2023;7(5) (no pagination)doi:https://dx.doi.org/10.1016/j.rpth.2023.100192

44. Mei H, Liu X, Li Y, et al. A multicenter, randomized phase III trial of hetrombopag: a novel thrombopoietin receptor agonist for the treatment of immune thrombocytopenia. *Journal of hematology & oncology*. 2021;14(1):37. doi:https://dx.doi.org/10.1186/s13045-021-01047-9

45. Liu Q, Shen Y, Li Y, et al. Efficiency and safety of eltrombopag for multi-line failed Chinese patients with immune thrombocytopenia: cases with decreased megakaryocyte response well from single-center experience. *Immunologic Research*. 2022;70(1):67-74. doi:https://dx.doi.org/10.1007/s12026-021-09245-w

46. Newland A, Godeau B, Priego V, et al. Remission and platelet responses with romiplostim in primary immune thrombocytopenia: final results from a phase 2 study. *British journal of haematology*. 2016;172(2):262-73. doi:https://dx.doi.org/10.1111/bjh.13827

47. Lucchini E, Palandri F, Volpetti S, et al. Eltrombopag second-line therapy in adult patients with primary immune thrombocytopenia in an attempt to achieve sustained remission off-treatment: results of a phase II, multicentre, prospective study. *British Journal of Haematology*. 2021;193(2):386-396. doi:https://dx.doi.org/10.1111/bjh.17334

48. Schifferli A, Rufer A, Rovo A, et al. Immunomodulation with romiplostim as a second-line strategy in primary immune thrombocytopenia: The iROM study. *British Journal of Haematology*. 2023;203(1):119-130. doi:https://dx.doi.org/10.1111/bjh.19074

49. Nelson VS, Amini SN, Netelenbos T, et al. The 'Stop TPO-RA in ITP Patients' study: Clinical and immune modulatory effects of romiplostim tapering. *British journal of haematology*. 2025;206(6):1743-1753. doi:https://dx.doi.org/10.1111/bjh.20100

50. Cooper N, Ghanima W, Vianelli N, et al. Sustained response off-treatment in eltrombopag-treated adult patients with ITP who are refractory or relapsed after first-line steroids: Primary, final, and ad-hoc analyses of the Phase II TAPER trial. *American journal of hematology*. 2024;99(1):57-67. doi:https://dx.doi.org/10.1002/ajh.27131

51. Oka S, Ono K, Nohgawa M. The association of lymphocyte phenotypes and outcomes after discontinuing eltrombopag in immune thrombocytopenia. *International Journal of Clinical Practice*. 2021;75(5):e14057. doi:https://dx.doi.org/10.1111/ijcp.14057

52. George JN, Mathias SD, Go RS, et al. Improved quality of life for romiplostim-treated patients with chronic immune thrombocytopenic purpura: results from two randomized, placebo-controlled trials. Multicenter Study

Randomized Controlled Trial

Research Support, Non-U.S. Gov't. *British Journal of Haematology*. Feb 2009;144(3):409-15. doi:https://dx.doi.org/10.1111/j.1365-2141.2008.07464.x

53. Kuter DJ, Mathias SD, Rummel M, et al. Health-related quality of life in nonsplenectomized immune thrombocytopenia patients receiving romiplostim or medical standard of care. Comparative Study

Multicenter Study

Randomized Controlled Trial

Research Support, Non-U.S. Gov't. *American Journal of Hematology*. May 2012;87(5):558-61. doi:https://dx.doi.org/10.1002/ajh.23163

54. Mei H, Liu X, Li Y, et al. Dose tapering to withdrawal stage and long-term efficacy and safety of hetrombopag for the treatment of immune thrombocytopenia: Results from an open-label extension study. Clinical Trial, Phase III

Multicenter Study

Randomized Controlled Trial

Research Support, Non-U.S. Gov't. *J Thromb Haemost*. 03 2022;20(3):716-728. doi:https://dx.doi.org/10.1111/jth.15602

55. Li Z, Mou W, Lu G, et al. Low-dose rituximab combined with short-term glucocorticoids up-regulates Treg cell levels in patients with immune thrombocytopenia. *International journal of hematology*. 2011;93(1):91-8. doi:https://dx.doi.org/10.1007/s12185-010-0753-z

56. Ghanima W, Khelif A, Waage A, et al. Rituximab as second-line treatment for adult immune thrombocytopenia (the RITP trial): a multicentre, randomised, double-blind, placebo-controlled trial. *Lancet*. 2015;385(9978):1653-61. doi:https://dx.doi.org/10.1016/S0140-6736(14)61495-1

57. Arnold DM, Heddle NM, Carruthers J, et al. A pilot randomized trial of adjuvant rituximab or placebo for nonsplenectomized patients with immune thrombocytopenia. *Blood*. 2012;119(6):1356-62. doi:https://dx.doi.org/10.1182/blood-2011-08-374777

58. Godeau B, Porcher R, Fain O, et al. Rituximab efficacy and safety in adult splenectomy candidates with chronic immune thrombocytopenic purpura: results of a prospective multicenter phase 2 study. *Blood*. 2008;112(4):999-1004. doi:https://dx.doi.org/10.1182/blood-2008-01-131029

59. Boccia R, Cooper N, Ghanima W, et al. Fostamatinib is an effective second-line therapy in patients with immune thrombocytopenia. *British Journal of Haematology*. 2020;190(6):933-938. doi:https://dx.doi.org/10.1111/bjh.16959

60. Gonzalez-Lopez TJ, Bermejo-Vega N, Cardesa-Cabrera R, et al. Fostamatinib effectiveness and safety for immune thrombocytopenia in clinical practice. *Blood*. 2024;144(6):646-656. doi:https://dx.doi.org/10.1182/blood.2024024250

61. Moore DC, Elmes JB, Arnall JR, Pineda-Roman M. Real-world clinical outcomes with fostamatinib for the treatment of refractory chronic immune thrombocytopenia: a single-center experience. *Blood Coagulation & Fibrinolysis*. 2024;35(6):316-320. doi:https://dx.doi.org/10.1097/MBC.0000000000001319

62. Jimenez-Barcenas R, Garcia-Donas-Gabaldon G, Campos-Alvarez RM, et al. Treatment with fostamatinib in patients with immune thrombocytopenia: Experience from the Andalusian region in Spain-The Fostasur Study. *British Journal of Haematology*. 2024;204(5):1977-1985. doi:https://dx.doi.org/10.1111/bjh.19443

63. Kuter DJ, Ghanima W, Cooper N, et al. Safety and efficacy of rilzabrutinib vs placebo in adults with immune thrombocytopenia: the phase 3 LUNA3 study. *Blood*. 2025;145(24):2914-2926. doi:https://dx.doi.org/10.1182/blood.2024027336

64. Yan S, Zhou H, Huang R, et al. A phase 2 trial of orelabrutinib showing promising efficacy and safety in patients with persistent or chronic primary immune thrombocytopenia. *American journal of hematology*. 2024;99(7):1392-1395. doi:https://dx.doi.org/10.1002/ajh.27303

65. Kuter DJ, Mayer J, Efraim M, et al. Long-term treatment with rilzabrutinib in patients with immune thrombocytopenia. *Blood Advances*. 2024;8(7):1715-1724. doi:https://dx.doi.org/10.1182/bloodadvances.2023012044

66. Huang QS, Fu HX, Wang CC, et al. Efficacy and Safety of the Bruton's Tyrosine Kinase Inhibitor Zanubrutinib in Immune Thrombocytopenia. *American Journal of Hematology*. 2025;doi:https://dx.doi.org/10.1002/ajh.27718

67. Su J, Xu M, Dong Z, et al. Efficacy and safety of azathioprine plus prednisone versus prednisone alone as first-line treatment for antinuclear antibody-positive immune thrombocytopenia: a retrospective cohort study. *Hematology*. 2023;28(1):2196864. doi:https://dx.doi.org/10.1080/16078454.2023.2196864

68. Abdallah GEM, Elbiih EAS, Sayed D, Moeen SM, Gafer S, Thabet AF. Revisiting the management of chronic ITP; a randomized controlled clinical trial. *Platelets*. 2021;32(2):243-249. doi:https://dx.doi.org/10.1080/09537104.2020.1738367

69. Chang H, Tang TC, Hung YS, et al. Immune thrombocytopenia: Effectiveness of frontline steroids and comparison of azathioprine, splenectomy, and rituximab as second-line treatment. *European Journal of Haematology*. 2018;101(4):549-555. doi:https://dx.doi.org/10.1111/ejh.13144

70. Mishra K, Pramanik S, Sandal R, et al. Safety and efficacy of azathioprine in immune thrombocytopenia. *American Journal of Blood Research*. 2021;11(3):217-226.

71. Cuker A, Despotovic JM, Grace RF, et al. Tapering thrombopoietin receptor agonists in primary immune thrombocytopenia: Expert consensus based on the RAND/UCLA modified Delphi panel method. *Res Pract Thromb Haemost*. Jan 2021;5(1):69-80. doi:10.1002/rth2.12457

72. Amini SN, Nelson VS, Sobels A, Schoones JW, Zwaginga JJ, Schipperus MR. Autologous platelet scintigraphy and clinical outcome of splenectomy in immune thrombocytopenia: A systematic review and meta-analysis. *Crit Rev Oncol Hematol*. Sep 2020;153:103040. doi:10.1016/j.critrevonc.2020.103040

73. González-Porras JR, Godeau B, Carpenedo M. Switching thrombopoietin receptor agonist treatments in patients with primary immune thrombocytopenia. *Ther Adv Hematol*. 2019;10:2040620719837906. doi:10.1177/2040620719837906

74. Provan D, Arnold DM, Bussel JB, et al. Updated international consensus report on the investigation and management of primary immune thrombocytopenia. *Blood Adv*. Nov 26 2019;3(22):3780-3817. doi:10.1182/bloodadvances.2019000812

75. Sanz M, Vicente García V, Fernández A, et al. [Guidelines for diagnosis, treatment and monitoring of primary immune thrombocytopenia]. *Med Clin (Barc)*. Mar 17 2012;138(6):261.e1-261.e17. Diagnóstico, tratamiento y seguimiento de la trombocitopenia inmune primaria. doi:10.1016/j.medcli.2011.11.011

76. Kashiwagi H, Kuwana M, Hato T, et al. Reference guide for management of adult immune thrombocytopenia in Japan: 2019 Revision. *Int J Hematol*. Mar 2020;111(3):329-351. doi:10.1007/s12185-019-02790-z

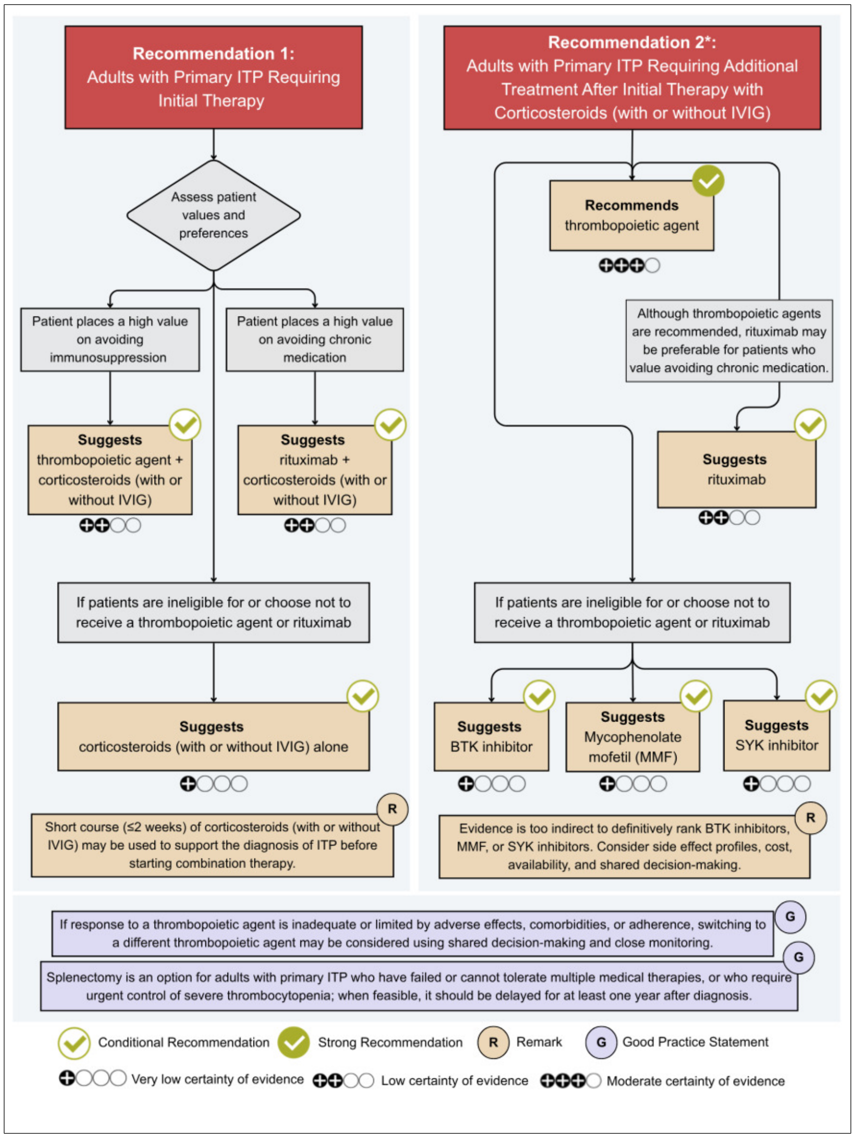
77. Chowdhury SR, Sirotich E, Guyatt GH, et al. Guideline on the emergency management of critical bleeding in patients with immune thrombocytopenia. *Blood Adv*. Feb 4 2026;doi:10.1182/bloodadvances.2025018818

78. Schünemann HJ, Wiercioch W, Brozek J, et al. GRADE Evidence to Decision (EtD) frameworks for adoption, adaptation, and de novo development of trustworthy recommendations: GRADE-ADOLOPMENT. *J Clin Epidemiol*. Jan 2017;81:101-110. doi:10.1016/j.jclinepi.2016.09.009

	Recommendation Strength	
	"Recommends…" / "Recommends against…"	"Suggests…" / "Suggests against…"
	Interpretation of Strong Recommendations	**Interpretation of Conditional Recommendations**
Patients	Most patients in this situation would want the recommended course of action, and only a small proportion would not.	Most patients in this situation would want the suggested course of action, but many would not. Decision aids may be useful in helping patients to make decisions consistent with their individual risks, values, and preferences.
Clinicians	Most clinicians should follow the recommended course of action. Formal decision aids are not likely to be needed to help individual patients make decisions consistent with their values and preferences.	Different choices will be appropriate for individual patients; clinicians must help each patient arrive at a management decision consistent with the patient's values and preferences. Decision aids may be useful in helping individuals to make decisions consistent with their individual risks, values, and preferences.
Policymakers	The recommendation can be adopted as policy in most situations. Adherence to this recommendation according to the guideline could be used as a quality criterion or performance indicator.	Policymaking will require substantial debate and involvement of various stakeholders. Performance measures should assess if decision making is appropriate.
Researchers	The recommendation is supported by credible research or other convincing judgments that make additional research unlikely to alter the recommendation. On occasion, a strong recommendation is based on low or very low certainty in the evidence. In such instances, further research may provide important information that alters the recommendations.	The recommendation is likely to be strengthened (for future updates or adaptation) by additional research. An evaluation of the conditions and criteria (and the related judgments, research evidence, and additional considerations) that determined the conditional (rather than strong) recommendation will help identify possible research gaps.

Evidence Certainty	
High Certainty	
⊕⊕⊕⊕	There is high confidence that the true value of the estimate of interest is on one side of a threshold of interest or within a specific range.
Moderate Certainty	
⊕⊕⊕◯	There is moderate confidence that that the true value of the estimate of interest is on one side of a threshold of interest or within a certain range. The true value of the estimate may deviate slightly from the target of the certainty rating (i.e. may possibly fall in a different range).
Low Certainty	
⊕⊕◯◯	There is low confidence that that the true value of the estimate of interest is on one side of a threshold of interest or within a certain range. The true value of the estimate may deviate from the target of the certainty rating (i.e. likely fall in a different range).
Very Low Certainty	
⊕◯◯◯	There is low confidence that that the true value of the estimate of interest is on one side of a threshold of interest or within a certain range. The true value of the estimate may deviate from the target of the certainty rating (i.e. likely fall in a different range).

Figure 3



Recommendation 1:
Adults with Primary ITP Requiring Initial Therapy
Assess patient values and preferences
Patient places a high value on avoiding immunosuppression
Patient places a high value on avoiding chronic medication
Suggests thrombopoietic agent + corticosteroids (with or without IVIG)
Suggests rituximab + corticosteroids (with or without IVIG)
If patients are ineligible for or choose not to receive a thrombopoietic agent or rituximab
Suggests corticosteroids (with or without IVIG) alone
R
Short course (≤2 weeks) of corticosteroids (with or without IVIG) may be used to support the diagnosis of ITP before starting combination therapy.
Recommendation 2*:
Adults with Primary ITP Requiring Additional Treatment After Initial Therapy with Corticosteroids (with or without IVIG)
Recommends thrombopoietic agent
Although thrombopoietic agents are recommended, rituximab may be preferable for patients who value avoiding chronic medication.
Suggests rituximab
If patients are ineligible for or choose not to receive a thrombopoietic agent or rituximab
Suggests BTK inhibitor
Suggests Mycophenolate mofetil (MMF)
Suggests SYK inhibitor
R
Evidence is too indirect to definitively rank BTK inhibitors, MMF, or SYK inhibitors. Consider side effect profiles, cost, availability, and shared decision-making.
G
If response to a thrombopoietic agent is inadequate or limited by adverse effects, comorbidities, or adherence, switching to a different thrombopoietic agent may be considered using shared decision-making and close monitoring.
G
Splenectomy is an option for adults with primary ITP who have failed or cannot tolerate multiple medical therapies, or who require urgent control of severe thrombocytopenia; when feasible, it should be delayed for at least one year after diagnosis.
Conditional Recommendation
Strong Recommendation
R Remark
G Good Practice Statement
Very low certainty of evidence
Low certainty of evidence
Moderate certainty of evidence